



legoESM v0.1: A Differentiable Earth System Model in JAX

Technical Documentation of Equations and Implementations

legoESM Development Team

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Contents

1	Introduction	6
2	Model Architecture	6
3	Grid Systems	7
3.1	Cubed-Sphere Grid	7
3.1.1	Face-to-Cartesian mapping	7
3.1.2	Cell area	9
3.1.3	Grid spacing	9
3.1.4	Grid angle and wind rotation	9
3.1.5	Halo exchange	10
3.2	Gaussian Grid and Spectral Transforms	10
3.2.1	Grid dimensions	10
3.2.2	Associated Legendre polynomials	11
3.2.3	Spherical harmonic transforms	11
3.2.4	Spectral operators	11
3.2.5	Velocity from vorticity and divergence	11
3.3	Sigma Vertical Coordinate	11
3.3.1	Simmons–Burridge geopotential	11
3.3.2	Sigma-dot	12
3.3.3	Pressure velocity	12
3.4	Height and z^* Coordinates	12
4	Core Operators	12
4.1	Gradient	12
4.2	Divergence (flux form)	12
4.3	Vorticity	13
4.4	C-D Grid Operators (FV3-style, <code>operators_cdgrid.py</code>)	13
4.5	Laplacian and Hyperdiffusion	14
5	Finite-Volume Transport	14
5.1	MUSCL Reconstruction (2nd order, <code>halo = 1</code>)	14
5.2	PPM Reconstruction (4th order, <code>halo = 2</code>)	14
5.2.1	Fourth-order edge values	14
5.2.2	Monotonicity constraint	15
5.2.3	Colella–Woodward limiting	15

5.3	WENO5 Reconstruction (5th order, halo = 2)	15
5.3.1	Left-biased reconstruction at $i + \frac{1}{2}$	15
5.3.2	Smoothness indicators	15
5.3.3	Nonlinear weights	15
5.3.4	Right-biased reconstruction	15
5.4	Rusanov Flux	16
5.5	Well-balanced scheme	16
6	FC-Gram Spectral Operators	16
6.1	Extension operator	16
6.2	Spectral differentiation	16
6.3	C-grid-style divergence damping	16
7	Atmospheric Dynamics	17
7.1	Shallow Water Equations	17
7.2	Hydrostatic Primitive Equations (σ -coordinates)	17
7.3	Spectral Primitive Equations (Vorticity–Divergence)	18
7.4	Non-Hydrostatic Compressible Euler Equations	18
7.4.1	Split-explicit time integration	18
7.4.2	Sponge layer	19
7.4.3	Turbulence closure: CRM, LES, and DNS	19
8	Atmospheric Physics	19
8.1	Gray Radiation (Frierson 2006)	19
8.1.1	Longwave optical depth	20
8.1.2	Two-stream LW fluxes	20
8.1.3	Shortwave (Beer–Lambert)	20
8.1.4	Heating rate	20
8.2	RRTMGP Correlated- k Radiation	20
8.3	SBM Convection (Frierson 2007)	20
8.3.1	Moist adiabatic lapse rate	21
8.3.2	Saturation mixing ratio (Tetens)	21
8.3.3	CAPE and trigger	21
8.3.4	Enthalpy-conserving relaxation	21
8.3.5	Precipitation	21
8.4	Held–Suarez Forcing	21
8.4.1	Equilibrium temperature	21
8.4.2	Newtonian relaxation	21
8.4.3	Rayleigh friction	22
8.5	Louis (1979) Turbulence	22
8.5.1	Mixing length	22
8.5.2	Richardson number	22
8.5.3	Stability functions	22
8.6	Vertical Diffusion (Implicit Thomas Algorithm)	22
8.7	Surface Fluxes	22
8.7.1	Constant coefficients (default)	23
8.7.2	Monin–Obukhov Similarity Theory (MOST)	23
8.7.3	COARE 3.0 roughness (Fairall et al. 2003)	23
8.7.4	Large & Yeager (2004) roughness	24
8.8	Additional Turbulence Schemes	24
8.8.1	Smagorinsky	24
8.8.2	YSU (Yonsei University)	24

8.8.3	CLUBB-lite	24
8.8.4	EDMF Turbulence	24
8.9	Additional Convection Schemes	24
8.9.1	Kuo (1974)	24
8.9.2	Mass-Flux Convection (Arakawa–Wu Type)	24
8.9.3	EDMF Convection	24
8.9.4	Deep Convective Adjustment (DCA)	24
8.10	Gravity Wave Drag	25
8.11	Cloud Microphysics	25
9	Ocean Dynamics	25
9.1	Boussinesq Hydrostatic Primitive Equations	25
9.1.1	Tracer equations (skew-symmetric)	25
9.1.2	Free-surface equation	25
9.2	Barotropic Solver (Forward–Backward, Semi-Implicit Coriolis)	26
9.3	Equation of State (Wright 1997)	26
9.3.1	Linear Equation of State	26
9.4	Hydrostatic Pressure and Vertical Velocity	26
9.5	Ocean Vertical Mixing	27
9.5.1	K-Profile Parameterisation (KPP, Large et al. 1994)	27
9.5.2	Richardson-Number Mixing	27
9.5.3	Constant Mixing	27
9.6	Ocean Lateral Mixing	27
9.6.1	Harmonic (Laplacian) Diffusion	27
9.6.2	Biharmonic (∇^4) Diffusion	27
9.6.3	Gent–McWilliams / Redi (GM/Redi)	27
9.7	Ocean Bottom Drag	27
9.8	Ocean Surface Forcing	28
9.9	Shortwave Penetration	28
9.10	Ocean Convection	28
9.10.1	Enhanced Diffusion	28
9.10.2	Plume Model	28
9.11	Ocean Conservation Fixers	28
9.12	Tracer Advection Schemes	28
9.13	Realistic-Geometry Spinup	29
10	Time Integration	29
10.1	SSP-RK3 (Shu–Osher 1988)	29
10.2	Robert-Asselin-Williams (RAW) Time Filter (v3.7)	29
10.3	Adaptive Hyperdiffusion (v3.7)	29
10.4	Split-Explicit Methods	30
11	Land, Ice, and Coupler	30
11.1	Slab Land Model	30
11.1.1	Energy balance	30
11.1.2	Bucket hydrology	30
11.2	Sea-Ice Model	30
11.2.1	Surface temperature	30
11.2.2	Growth and melt	30
11.2.3	Concentration	30
11.2.4	EVP Rheology (Hunke & Dukowicz 1997)	30
11.2.5	Multi-category ice	31

11.3	Land Carbon Cycle and Stomatal Conductance	31
11.3.1	DALEC-990 Carbon Pools	31
11.3.2	Farquhar (1980) Photosynthesis	31
11.3.3	Stomatal conductance	31
11.4	Plant Functional Types (PFTs)	31
11.5	DifferLand Training	32
11.6	Tile Coupler	32
12	Machine Learning: Spherical Fourier Neural Operator	32
12.1	Architecture	33
12.2	Spectral convolution	33
12.3	Neural and Learned Physics	33
12.3.1	NeuralPhysics	33
12.3.2	LearnedColumn	33
13	Conservation Corrections	33
13.1	Atmosphere	33
13.2	Ocean	34
14	Physical Constants	34
15	Ocean Biogeochemistry	34
15.1	Carbonate Chemistry and Air-Sea CO ₂ Exchange	34
15.2	NPZD Ecosystem Model	34
16	CMIP Infrastructure	35
16.1	CF/CMOR-Compliant Output	35
16.2	AMIP CMIP6 Forcing Deck	35
16.3	Experiment Templates	35
16.4	Restart and Reproducibility	35
16.5	Tuning Guide	36
16.6	OMIP Simulation Pipeline	36
17	4D-Var Data Assimilation (v3.9)	36
17.1	Cost Function	36
17.2	Control Vector	36
17.3	Background Error Covariance	36
17.4	Observation Operators	37
17.5	Minimiser	37
17.6	DA Cycling and Incremental Analysis	37
18	References	37

Introduction

legoESM is a modular Earth System Model written entirely in JAX, supporting automatic differentiation through all components. The model couples atmospheric, oceanic, land, sea-ice, and biogeochemical components on the sphere, with 25+ dynamical cores spanning five discretisation families (C-D grid FV3, finite-volume PPM, pseudospectral, FC-Gram, C-grid) and 35+ physics parameterisations. The cubed-sphere implementation is unified around a single FV3-style C-D grid discretisation (Lin 2004, Putman & Lin 2007) shared by atmosphere (shallow water, hydrostatic PE, non-hydrostatic CE) and ocean, eliminating the Hollingsworth–Kallberg instability. The cubed-sphere transport uses an FV3-faithful PPM reconstruction with monotonicity limiters and flux-form mass transport on staggered grids, complemented by an FV3-exact forward–backward shallow-water core (`fv3_sw_core.py`) implementing the full `c_sw/d_sw` stepping and `d2a2c_vect` (D-grid $\rightarrow$ A-grid $\rightarrow$ C-grid) vector conversion with duo-grid sub-grid metrics ($\sin \alpha_{sg}$, $\cos \alpha_{sg}$) following Mouallem et al. (2023). The model includes a land carbon cycle with Farquhar photosynthesis and stomatal conductance, ocean biogeochemistry (abiotic carbon + NPZD ecosystem), sea-ice dynamics with EVP rheology and multi-category ice, CMIP6-class infrastructure (CF/CMOR output across four tables, experiment templates, restart reproducibility, documented tuning), and a fully differentiable 4D-Var data assimilation system leveraging JAX autodiff for adjoint-free gradient computation. The MPI-distributed runtime supports native 4D halo exchange (one message per neighbour for all vertical levels), reverse-mode AD through MPI via a custom `sendrecv` VJP wrapper, and AD-safe `allreduce(SUM)` for gradient-compatible global reductions. The codebase comprises approximately 385 source modules ($\sim 102,000$ lines of code) validated by 3489 tests across 216 test files.

Design principles.

- Pure-JAX implementation: all components are compatible with `jit`, `grad`, and `vmap`.
- Modular physics: each parameterisation follows a factory pattern returning a callable `physics_fn(state, grid, coord)  $\rightarrow$  Tendencities`.
- Multiple dynamical cores share the same physics interface, enabling interchangeable experiments.
- Neural-network components (SFNO) integrate seamlessly via the same JAX computational graph.

Model Architecture

legoESM is organised as a *federation* of eight independently installable packages that all ship into one shared `legoesm` PEP-420 namespace, layered as an acyclic dependency graph whose import boundaries are enforced by zero-ignore `lint-imports` contracts (Fig. 1). A single `legoesm-core` substrate provides fields, grids, vertical coordinates, operators, the device/precision runtime, the parallel halo/reduction layer, and the grid $\times$ complexity capability matrix; the four Earth-system components (`atmosphere`, `ocean`, `land`, `ice`) depend *only* on the substrate and never on each other, so each can be `pip` installed and gradient-checked in isolation. Above them sits an orchestration cluster — `coupler+driver`, `ml+training+da`, and `tools` — whose mutual edges form a uv-workspace cycle, so the three install together.

The “lego” in legoESM is the second design axis: standard tensor-in / tendency-out interfaces let every block — dynamical core, each physics parameterisation, grid, vertical coordinate, time integrator, and even classical-versus-neural physics — be swapped independently by a string scheme name resolved through a factory (Table 1). Figure 2 traces one coupled time step through these swap points.

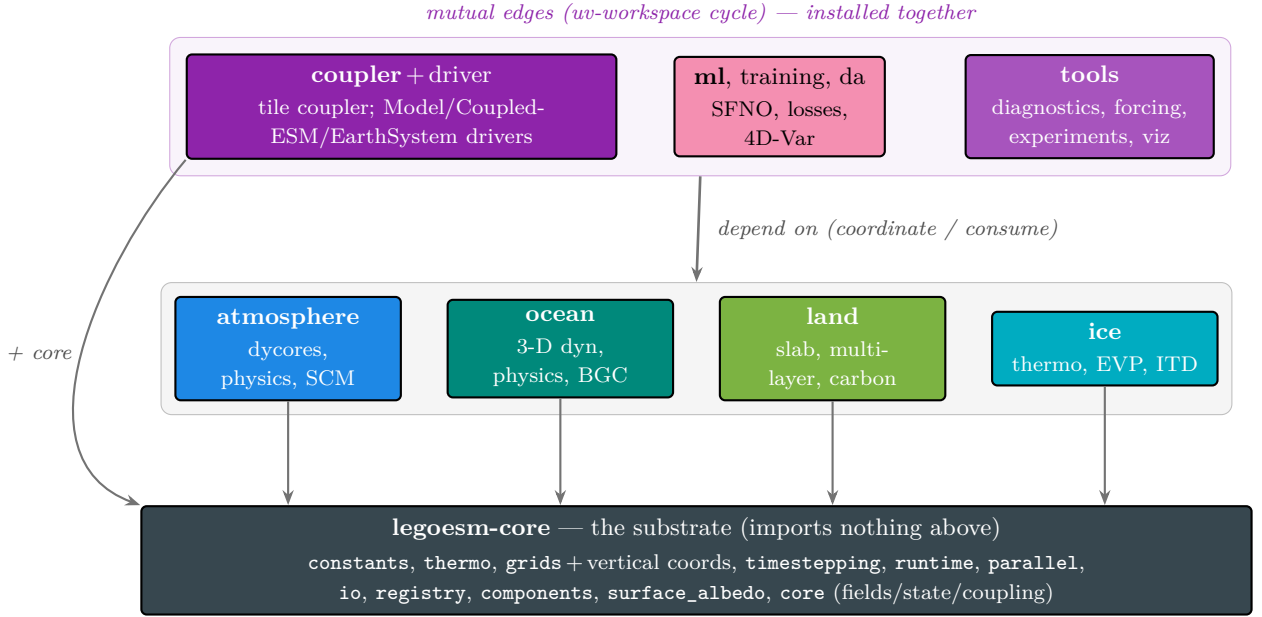


Figure 1: **Federation dependency DAG**. Arrows point from a package to what it imports (“depends on”); the layering is gated by zero-ignore import contracts (see **FEDERATION.md**). The four components are mutually independent and standalone-installable (contract #2); the orchestration cluster (coupler / ml / tools) sits above and carries the only mutual edges.

A coupled step runs at three nested cadences. The host *segment loop* (`build_segment_fn` → `run_segment_jit`) drives a `jax.lax.scan` over inner atmosphere steps; each inner step advances the dycore (u, v, T, p_s) , applies the gated dry-mass fixer, runs the unified physics pipeline (radiation sub-cycled by `lax.cond`; then convection, microphysics, surface exchange, turbulence/BL, and gravity-wave drag summed as tendencies), and closes with the moisture/energy state update. After each segment, `_segment_hook` sub-cycles the slow components at `coupling_dt` (default 3600 s): the ocean step, the tile surface coupler (area-weighted bulk-flux blend over ocean/ice/land/lake), and the gated CO_2 tracer, followed by conservation diagnostics. Because every block is a pure JAX pytree function and all communication is differentiable, the same graph supports forward simulation, `jit` compilation, and reverse-mode `grad`.

Grid Systems

Cubed-Sphere Grid

Six faces of a cube are mapped to the sphere via **gnomonic (central) projection** (Ronchi et al., 1996; Putman and Lin, 2007). This quasi-uniform tiling replaces the polar singularity and meridian convergence of a latitude–longitude mesh with six logically rectangular panels, so the maximum stable timestep is set by a near-uniform grid spacing rather than by the vanishing zonal spacing at the poles. Let $(\alpha_x, \alpha_y) \in [-\pi/4, \pi/4]^2$ be the equiangular gnomonic coordinates on a face; because great circles map to straight lines in (α_x, α_y) , the projection is naturally suited to conservative finite-volume discretisation (Sadourny, 1972).

Face-to-Cartesian mapping

For face 0 (+x face), the gnomonic map sends the planar face coordinates to a ray from the cube centre:

$$(x, y, z) = (1, \tan \alpha_x, \tan \alpha_y), \quad (1)$$

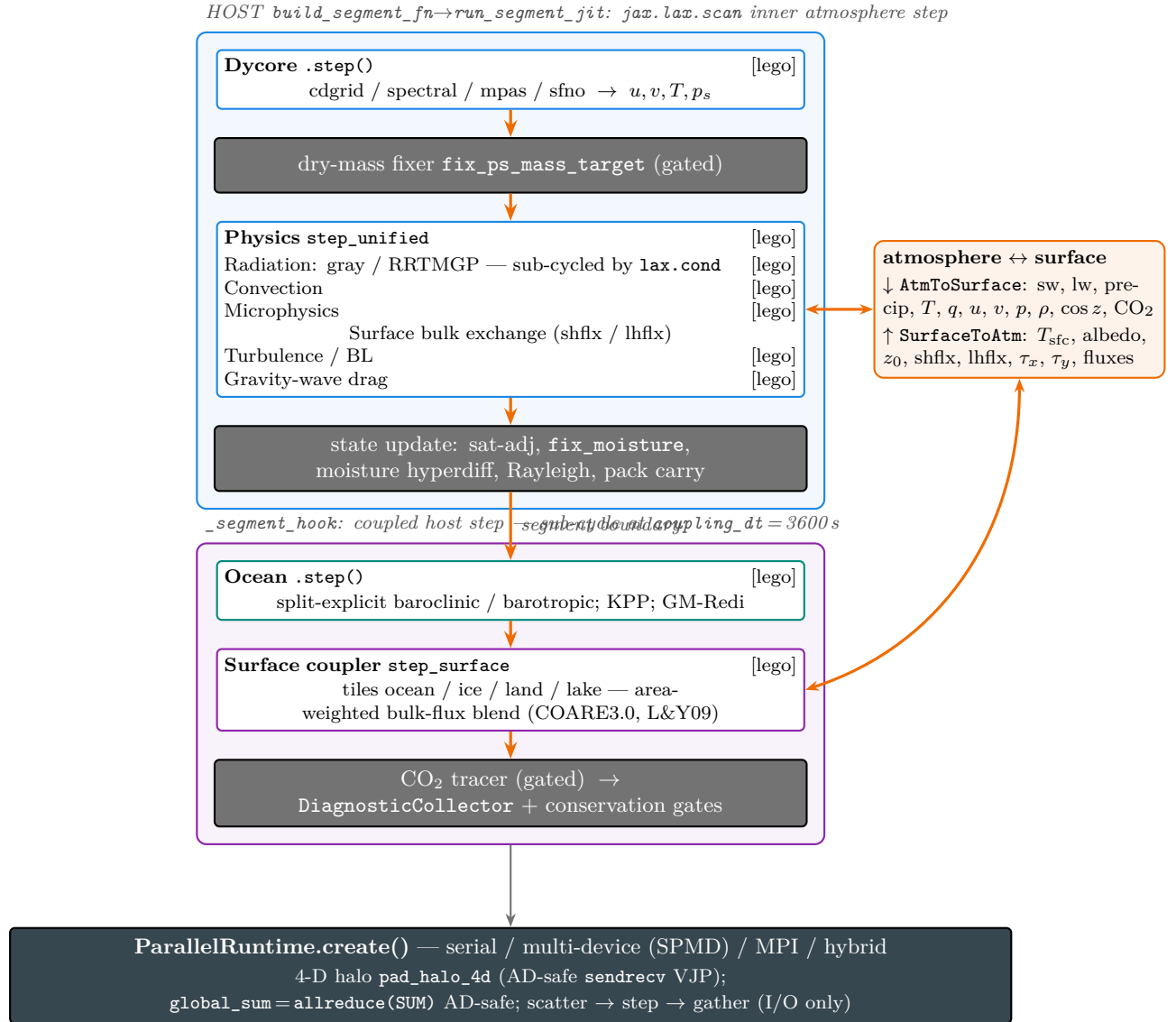


Figure 2: **Coupled time-step data flow.** White blocks tagged [lego] are factory-swapped by scheme name (Table 1); dark blocks are fixed numerics. The atmosphere inner step (blue) runs inside a `jax.lax.scan`; the slow components (purple) sub-cycle at `coupling_dt`. The orange link is the bidirectional atmosphere–surface flux exchange. The entire graph executes on one `ParallelRuntime` with AD-safe halo exchange and `allreduce(SUM)` reductions, so `jax.grad` flows end-to-end through the coupled step.

Table 1: **The “lego” swap dials.** Each axis is selected independently by a string scheme name resolved through the listed factory; an unknown name raises **ValueError**. Menus abbreviated.

Swap axis	Factory entry point	Example menu
Horizontal grid	<code>create_grid()</code>	cubed-sphere, Gaussian/spectral, lat-lon, MPAS Voronoi, tripolar (ocean)
Dynamical core	<code>create_atmosphere_dycore()</code>	{cdgrid, spectral, mpas, sfno} × {shallow water, hydrostatic PE, NH compressible Euler}; lat-lon C-grid; plane
Vertical coordinate	<code>create*_coordinate()</code>	sigma, hybrid σ - p , height / z^*
Time integrator	<code>dispatch_integrator()</code>	ssp_rk3, ssp_rk3_scan, ssp_rk34, ssp_rk54, rk4
Radiation	<code>make_radiation_physics()</code>	gray (Frierson), RRTMG correlated- k
Convection	<code>make_convection_physics()</code>	Bechtold, Tiedtke, Kain-Fritsch, Zhang-McFarlane, Emanuel, Kuo, SBM
Microphysics	<code>make_microphysics_physics()</code>	Kessler, Sundqvist, Morrison, Seifert-Beheng, Thompson, ML
Turbulence / BL	<code>make_turbulence_physics()</code>	Louis, Holtslag-Boville, YSU, EDMF, TKE, CLUBB-lite, Smagorinsky
Gravity-wave drag	<code>make_gwd_physics()</code>	Lindzen, Hines, McFarlane, Rayleigh, spectral, ML
Physics core	<code>make_{physics,hybrid,neural}</code>	classical, hybrid, neural (SFNO)
Ocean vertical mixing	<code>make_vertical_mixing_physics()</code>	(KPP, Richardson, constant, Jayne-St-Laurent tidal
Ocean lateral mixing	<code>make_lateral_mixing_physics()</code>	harmonic, biharmonic, GM-Redi, Visbeck, Leith, backscatter
Ocean surface forcing	<code>make_surface_forcing_physics()</code>	(bulk, restoring, prescribed
Complexity ladder	<code>resolve_model_complexity()</code>	idealized, intermediate, full

with analogous expressions for the remaining five faces obtained by axis permutation and sign. Points are normalised to the unit sphere: $\hat{\mathbf{r}} = \mathbf{r}/|\mathbf{r}|$, and the geographic longitude λ and latitude ϕ follow as $\lambda = \text{atan2}(\hat{y}, \hat{x})$, $\phi = \arcsin(\hat{z})$.

Cell area

The area element on the gnomonic equidistant grid is

$$\Delta A = \frac{R^2 \Delta \alpha^2}{\cos^2 \alpha_x \cos^2 \alpha_y D^3}, \quad D = \sqrt{1 + \tan^2 \alpha_x + \tan^2 \alpha_y}, \quad \Delta \alpha = \frac{\pi}{2n}. \quad (2)$$

Here R is the planetary radius, $\Delta \alpha$ the uniform angular cell width on a C_n face of $n \times n$ cells, and D the distance from the cube centre to the projected point. The D^{-3} factor is exactly the Jacobian of the gnomonic map; summing ΔA over the six faces recovers $4\pi R^2$, the discrete area-conservation property the finite-volume operators depend on.

Grid spacing

Great-circle distances use the chord-to-arc formula:

$$d = R \cdot 2 \arcsin\left(\frac{|\mathbf{r}_1 - \mathbf{r}_2|}{2}\right). \quad (3)$$

Metric factors $\Delta x_{i,j}$ and $\Delta y_{i,j}$ are computed as two-cell spans for second-order centred differences.

Grid angle and wind rotation

The angle θ between the grid x -axis and geographic east is

$$\theta = \text{atan2}\left(\frac{\partial \phi}{\partial i}, \frac{\partial \lambda}{\partial i} \cos \phi\right). \quad (4)$$

Wind rotation between geographic (u_E, v_N) and grid-aligned (u_g, v_g) :

$$\begin{pmatrix} u_E \\ v_N \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} u_g \\ v_g \end{pmatrix}. \quad (5)$$

Halo exchange

Scalar fields are exchanged directly across the six-face connectivity table. Vector fields are first rotated to geographic components, padded as scalars, then rotated back using the padded grid angle, because the grid-aligned basis is discontinuous across a panel edge and a naive copy of (u_g, v_g) would inject a spurious rotation at the seam. A discrete duo-grid treatment of the panel edges removes the residual edge imprint that a single-grid halo leaves on differential operators (Mouallem et al., 2023).

Two halo widths are supported. **Halo = 1** (one ghost cell per side) suffices for second-order centred differences and MUSCL reconstruction. **Halo = 2** (two ghost cells per side) is required by higher-order reconstruction stencils (PPM, WENO5). The halo exchange interpolates neighbour-face strips to the correct gnomonic position at each depth, removing the $O(\Delta\alpha)$ mismatch near cube corners.

Under MPI, the native **4D halo exchange** path (`pad_halo_4d`, `pad_halo_vector_4d`) exchanges all vertical levels in a single MPI message per neighbour, replacing the earlier per-level `vmap(pad_halo)` path. All `sendrecv` calls go through a `@jax.custom_vjp` wrapper that swaps source/dest in the backward pass, enabling `jax.grad` through distributed halo exchange for differentiable MPI simulations.

The May 2026 GPU-scaling branch (#228) further compresses per-RK3-step collectives across all dycore backends: cubed-sphere allreduces fuse $4 \rightarrow 1$ (halo bandwidth reduced $\sim 36\text{--}144\times$ at high resolution), the spectral PE FFT and Legendre transforms fuse into a single pass with stacked per-tracer SH analyses, and MPAS `sendrecvs` drop $12 \rightarrow 6$ with mass-fix payloads $3 \rightarrow 2$. Fifteen SPMD-correctness unit tests gate the changes. Real-hardware Levante-cluster validation is described in `docs/REAL_HARDWARE_SCALING.md` and `docs/GPU_SCALING_BRANCH.md`.

Gaussian Grid and Spectral Transforms

The spectral dynamical cores represent fields by their spherical-harmonic coefficients and use a Gaussian grid with triangular truncation at total wavenumber $n_{\max}$ (Bourke, 1972). Triangular truncation is chosen because it is isotropic: the retained set of harmonics is invariant under rotation of the sphere, so no direction is preferentially resolved.

Grid dimensions

$$n_{\text{lat}} = \lceil \frac{3}{2}(n_{\max} + 1) \rceil_{\text{even}}, \quad n_{\text{lon}} = 2 n_{\text{lat}}, \quad n_{\text{sh}} = \frac{(n_{\max} + 1)(n_{\max} + 2)}{2}. \quad (6)$$

The factor $\frac{3}{2}$ is the standard transform-grid choice that makes the Gaussian quadrature exact for the quadratic (advective) products, eliminating spectral aliasing. Latitudes are Gaussian quadrature points $\mu_k = \sin \phi_k$ with weights w_k ; longitudes are equispaced so the zonal direction can be treated by FFT.

Associated Legendre polynomials

Fully normalised (4π) polynomials $P_n^m(\mu)$ satisfy $\int |Y_n^m|^2 dA = 1$. They are computed via the recursion

$$P_m^m = -\sqrt{\frac{2m+1}{2m}} \sqrt{1-\mu^2} P_{m-1}^{m-1}, \quad (7)$$

$$P_n^m = \sqrt{\frac{4n^2-1}{n^2-m^2}} \left(\mu P_{n-1}^m - \sqrt{\frac{(n-1)^2-m^2}{4(n-1)^2-1}} P_{n-2}^m \right). \quad (8)$$

The derivative operator $H_n^m = -(1-\mu^2) dP_n^m/d\mu$ is used for meridional gradients.

Spherical harmonic transforms

Analysis (grid $\rightarrow$ spectral).

$$\hat{f}_{nm} = 2\pi \sum_k w_k P_n^m(\mu_k) \tilde{f}_m(\mu_k), \quad \tilde{f}_m = \frac{1}{n_{\text{lon}}} \text{FFT}[f]_m. \quad (9)$$

Synthesis (spectral $\rightarrow$ grid).

$$f(\phi, \lambda) = \text{iFFT} \left[\sum_{n=m}^{n_{\text{max}}} \hat{f}_{nm} P_n^m(\mu) \right]. \quad (10)$$

Spectral operators

Laplacian eigenvalue: $\nabla^2 \hat{f}_{nm} = -n(n+1)/a^2 \cdot \hat{f}_{nm}$. Inverse Laplacian: $\nabla^{-2} \hat{f}_{nm} = -a^2/[n(n+1)]$ for $n > 0$. Hyperdiffusion: $-\nu[n(n+1)/a^2]^p \hat{f}_{nm}$.

Velocity from vorticity and divergence

Streamfunction $\psi = \nabla^{-2}\zeta$, velocity potential $\chi = \nabla^{-2}\delta$. Velocity reconstruction:

$$u \cos \phi = \frac{\partial \psi}{\partial \theta} + \frac{\partial \chi}{\partial \lambda}, \quad v \cos \phi = \frac{\partial \psi}{\partial \lambda} - \frac{\partial \chi}{\partial \theta}, \quad (11)$$

where θ is colatitude.

Sigma Vertical Coordinate

$$\sigma = p/p_s, \quad \sigma \in [\sigma_{\text{top}}, 1]. \quad (12)$$

Half-level interfaces $\sigma_{k+1/2}$ are uniformly spaced; full levels $\sigma_k = (\sigma_{k-1/2} + \sigma_{k+1/2})/2$. Layer thickness $\Delta\sigma_k = \sigma_{k+1/2} - \sigma_{k-1/2}$.

Simmons–Burridge geopotential

The vertical finite differencing follows Simmons and Burridge (1981), constructed so that the discrete pressure-gradient and energy-conversion terms conserve total energy under adiabatic, frictionless flow. The geopotential at full level k is

$$\Phi_k = \Phi_s + R_d \sum_{k'=k+1}^{N-1} T_{k'} \ln \frac{\sigma_{k'+1/2}}{\sigma_{k'-1/2}} + \alpha_k R_d T_k, \quad (13)$$

where the Simmons–Burridge coefficient is

$$\alpha_k = 1 - \frac{\sigma_{k-1/2}}{\Delta\sigma_k} \ln \frac{\sigma_{k+1/2}}{\sigma_{k-1/2}}. \quad (14)$$

Sigma-dot

The vertical velocity in σ -space is obtained by integrating the continuity equation from the model top ($\dot{\sigma} = 0$):

$$\dot{\sigma}_{k+1/2} = \frac{\sigma_{k+1/2} - \sigma_{\text{top}}}{1 - \sigma_{\text{top}}} D_{\text{tot}} - \sum_{k'=0}^k \nabla \cdot \mathbf{v}_{k'} \Delta \sigma_{k'}, \quad D_{\text{tot}} = \sum_{k'=0}^{N-1} \nabla \cdot \mathbf{v}_{k'} \Delta \sigma_{k'}. \quad (15)$$

Pressure velocity

Since $p = \sigma p_s$, the material pressure tendency is

$$\omega \equiv \frac{dp}{dt} = \sigma \frac{dp_s}{dt} + p_s \dot{\sigma}, \quad (16)$$

where dp_s/dt is the *material* derivative of surface pressure (Eulerian tendency plus advection by the horizontal wind).

Height and z^* Coordinates

The terrain-following z^* coordinate:

$$z^* = H \frac{z - z_s}{H - z_s}, \quad J = \frac{\partial z}{\partial z^*} = \frac{H - z_s}{H}. \quad (17)$$

Physical height: $z(z^*, x, y) = z_s + z^* (H - z_s)/H$.

For the **ocean**, the z^* coordinate has a dynamic Jacobian:

$$J(\eta) = \frac{\eta + H}{H_{\text{max}}}, \quad (18)$$

where η is the free-surface elevation and H the resting depth.

Core Operators

All finite-difference operators use second-order centred differences on the A-grid (collocated) with halo padding on the cubed sphere. The discrete difference operators δ_i and δ_j are defined as centred differences of *cell-centre* values on the padded grid:

$$\delta_i[q]_{i,j} = q_{i+1,j} - q_{i-1,j}, \quad \delta_j[q]_{i,j} = q_{i,j+1} - q_{i,j-1}. \quad (19)$$

All metric-weighted fluxes ($u h_y$, $v h_x$) are evaluated at cell centres before differencing, consistent with the A-grid stencil.

Gradient

$$\left. \frac{\partial f}{\partial x} \right|_{i,j} = \frac{f_{i+1,j} - f_{i-1,j}}{2 \Delta x_{i,j}}, \quad \left. \frac{\partial f}{\partial y} \right|_{i,j} = \frac{f_{i,j+1} - f_{i,j-1}}{2 \Delta y_{i,j}}. \quad (20)$$

Divergence (flux form)

$$\nabla \cdot (u, v) = \frac{1}{2 A_{i,j}} \left[\delta_i(u h_y) + \delta_j(v h_x) \right], \quad (21)$$

where $h_x = \Delta x/2$, $h_y = \Delta y/2$ are half-edge lengths and A is the cell area.

Vorticity

$$\zeta = \frac{1}{2A_{i,j}} \left[\delta_i(v h_y) - \delta_j(u h_x) \right]. \quad (22)$$

C-D Grid Operators (FV3-style, `operators_cdgrid.py`)

The canonical cubed-sphere operator module uses FV3-style C-D grid staggering (Lin, 2004; Putman and Lin, 2007), shared by all atmosphere (SW, PE, CE) and ocean dynamical cores. D-grid winds at cell corners $(6, n+1, n+1)$ are prognostic; C-grid velocities at cell edges are diagnosed for mass/scalar transport. The C-D arrangement combines the favourable geostrophic adjustment of the C-grid with the cleanly defined corner vorticity of the D-grid.

FV3-faithful PPM transport (v3.9). The operator module was rewritten (~ 740 lines) with proper Piecewise Parabolic Method reconstruction (Colella and Woodward, 1984), Colella–Woodward monotonicity limiters, and flux-form mass transport on staggered grids, faithfully following the FV3 algorithm of Putman and Lin (2007). This replaces the earlier simplified A-grid transport with a fully staggered C-D grid transport that correctly handles cross-face halo exchange, geographic-to-grid wind rotation, and corner interpolation for the cubed sphere.

D-grid vorticity. Vorticity at each cell corner is the discrete circulation integral divided by the corner dual-cell area A_c :

$$\zeta_{\text{corner}} = \frac{1}{A_c} \oint \mathbf{u} \cdot d\mathbf{l} \approx \frac{1}{A_c} [u_{i,j}^D \Delta x - u_{i+1,j}^D \Delta x + v_{i,j+1}^D \Delta y - v_{i,j}^D \Delta y]. \quad (23)$$

Because vorticity is defined as the line integral of the tangential wind around the dual cell, the discrete curl inherits the Stokes-theorem identity exactly, so the kinetic-energy and enstrophy budgets remain mutually consistent. This circulation-based discretisation is exact on the D-grid and eliminates the Hollingsworth–Kallberg instability that plagues A-grid vorticity operators.

Arakawa–Lamb gradient. The Bernoulli (or Exner) gradient at D-grid corners is a 4-point Arakawa–Lamb average:

$$\left. \frac{\partial B}{\partial x} \right|_{i+\frac{1}{2}, j+\frac{1}{2}} = \frac{B_{i+1,j} + B_{i+1,j+1} - B_{i,j} - B_{i,j+1}}{2\Delta x}, \quad (24)$$

and analogously for $\partial B / \partial y$.

C-grid mass-flux divergence. Scalar transport uses upwind C-grid mass fluxes:

$$\frac{dh}{dt} = -\frac{1}{A} [(hu_c)_{i+\frac{1}{2},j} - (hu_c)_{i-\frac{1}{2},j} + (hv_c)_{i,j+\frac{1}{2}} - (hv_c)_{i,j-\frac{1}{2}}], \quad (25)$$

where u_c, v_c are diagnosed from u_d, v_d by face-edge averaging.

FV3 forward–backward shallow-water core (`fv3_sw_core.py`). An FV3-exact implementation of the complete `c_sw/d_sw` forward–backward stepping algorithm (Lin, 2004). The forward step (`c_sw`) computes C-grid mass fluxes and height tendencies; the backward step (`d_sw`) updates D-grid winds using the vorticity flux and pressure gradient. Key components:

- **d2a2c_vect:** D-grid $\rightarrow$ A-grid $\rightarrow$ C-grid vector conversion, faithfully porting the GFDL FV3 algorithm including edge-boundary stencils, one-sided interpolation near cube edges, and non-orthogonality corrections via $\cos \alpha / \sin^{-2} \alpha$ metrics.

- **PPM wind advection:** Piecewise Parabolic Method reconstruction for C-grid wind tendencies (`xtp_u`, `ytp_v`), with FV3-specific interpolation coefficients (A_1 , A_2 , C_1 – C_3).
- **PPM vorticity transport:** Monotonic PPM reconstruction for the absolute vorticity flux, stabilising the D-grid wind update.
- **Kinetic energy:** Cell-centred kinetic energy computed from A-grid velocities with optional 4th-order averaging via `a2b_ord4`.

Duo-grid sub-grid metrics (Mouallem et al., 2023). Following the duo-grid construction of Mouallem et al. (2023), the cubed-sphere C-D grid includes 9-point sub-grid angle metrics per cell, stored as $\sin \alpha_{\text{sg}}$ and $\cos \alpha_{\text{sg}}$ arrays of shape $(6, n, n, 9)$. These encode the local deviation from orthogonality at cell centres, edge midpoints, and corners, enabling FV3-faithful flux scaling at cube-face boundaries where the gnomonic projection introduces non-orthogonality. Additional C-D grid metrics include centre-to-centre distances Δx_c , Δy_c , and non-orthogonality factors $\cos \alpha_{\text{corner}}$, $(\sin^2 \alpha_{\text{corner}})^{-1}$ at D-grid corners.

Laplacian and Hyperdiffusion

$$\nabla^2 f = \nabla \cdot (\nabla f), \quad \text{Hyperdiffusion:} \quad -\nu \nabla^4 f = -\nu \nabla^2 (\nabla^2 f). \quad (26)$$

Finite-Volume Transport

The finite-volume dynamical cores use upwind reconstruction at cell edges coupled with a Rusanov approximate Riemann solver. The flux-form, dimensionally split transport follows the multidimensional flux-form semi-Lagrangian construction of Lin and Rood (1996) and the vertically Lagrangian finite-volume core of Lin (2004), which together guarantee discrete mass conservation. Three reconstruction methods are available, selected via the `limiter` parameter, trading order of accuracy against the width of the halo stencil.

MUSCL Reconstruction (2nd order, halo = 1)

Piecewise-linear reconstruction with slope limiting (van Leer, 1979). The limiter clips the reconstructed slope so the edge states never overshoot the neighbouring cell averages, preserving monotonicity. At the interface between cells i and $i+1$:

$$q_L = q_i + \frac{1}{2} \phi_i, \quad q_R = q_{i+1} - \frac{1}{2} \phi_{i+1}, \quad (27)$$

where ϕ_i is the limited slope. Two limiters are provided:

- **Minmod:** $\phi = \text{minmod}(\Delta q_+, \Delta q_-)$,
- **MC** (monotonised central): $\phi = \text{minmod}(\text{minmod}(2\Delta q_+, 2\Delta q_-), \frac{1}{2}(\Delta q_+ + \Delta q_-))$.

PPM Reconstruction (4th order, halo = 2)

The Piecewise Parabolic Method (Colella and Woodward, 1984) constructs a monotone parabola in each cell from two edge values and the cell average, giving fourth-order accuracy in smooth regions while suppressing oscillations at sharp gradients.

Fourth-order edge values

At the interface $i + \frac{1}{2}$:

$$\hat{q}_{i+1/2} = \frac{7}{12}(q_i + q_{i+1}) - \frac{1}{12}(q_{i-1} + q_{i+2}). \quad (28)$$

Monotonicity constraint

The edge value is clamped to lie between its two adjacent cell averages:

$$\hat{q}_{i+1/2} \leftarrow \text{median}(\hat{q}_{i+1/2}, q_i, q_{i+1}). \quad (29)$$

Colella–Woodward limiting

For each cell i with left edge value $a_L = \hat{q}_{i-1/2}$ and right edge value $a_R = \hat{q}_{i+1/2}$:

1. If $(a_R - q_i)(q_i - a_L) \leq 0$ (local extremum): set $a_L = a_R = q_i$.
2. Define $\delta_m = a_R - a_L$ and $\delta_6 = 6(q_i - \frac{1}{2}(a_L + a_R))$.
 - If $\delta_m \delta_6 > \delta_m^2$: set $a_L = 3q_i - 2a_R$.
 - If $\delta_m \delta_6 < -\delta_m^2$: set $a_R = 3q_i - 2a_L$.

The left and right states at each edge are then $q_L = a_R$ of the cell to the left and $q_R = a_L$ of the cell to the right.

WENO5 Reconstruction (5th order, halo = 2)

The 5th-order Weighted Essentially Non-Oscillatory scheme (Jiang and Shu, 1996) combines three 3rd-order sub-stencil polynomials with nonlinear weights that adapt to local smoothness, downweighting any stencil that straddles a discontinuity so that the reconstruction stays high-order in smooth flow yet essentially non-oscillatory near shocks.

Left-biased reconstruction at $i + \frac{1}{2}$

Using cells $q_{i-2}, \dots, q_{i+2}$:

$$q^{(0)} = \frac{1}{6}(2q_{i-2} - 7q_{i-1} + 11q_i), \quad d_0 = \frac{1}{10}, \quad (30)$$

$$q^{(1)} = \frac{1}{6}(-q_{i-1} + 5q_i + 2q_{i+1}), \quad d_1 = \frac{6}{10}, \quad (31)$$

$$q^{(2)} = \frac{1}{6}(2q_i + 5q_{i+1} - q_{i+2}), \quad d_2 = \frac{3}{10}. \quad (32)$$

Smoothness indicators

$$\beta_0 = \frac{13}{12}(q_{i-2} - 2q_{i-1} + q_i)^2 + \frac{1}{4}(q_{i-2} - 4q_{i-1} + 3q_i)^2, \quad (33)$$

$$\beta_1 = \frac{13}{12}(q_{i-1} - 2q_i + q_{i+1})^2 + \frac{1}{4}(q_{i-1} - q_{i+1})^2, \quad (34)$$

$$\beta_2 = \frac{13}{12}(q_i - 2q_{i+1} + q_{i+2})^2 + \frac{1}{4}(3q_i - 4q_{i+1} + q_{i+2})^2. \quad (35)$$

Nonlinear weights

$$\alpha_k = \frac{d_k}{(\varepsilon + \beta_k)^2}, \quad \omega_k = \frac{\alpha_k}{\sum_j \alpha_j}, \quad q_L = \sum_k \omega_k q^{(k)}, \quad \varepsilon = 10^{-6}. \quad (36)$$

Right-biased reconstruction

The right state q_R at $i + \frac{1}{2}$ is obtained by mirroring the stencil around the interface, using cells $q_{i-1}, \dots, q_{i+3}$ with ideal weights $d_0 = \frac{1}{10}$, $d_1 = \frac{6}{10}$, $d_2 = \frac{3}{10}$ applied to the reversed sub-stencils.

Rusanov Flux

At each edge, the mass flux is

$$F = \frac{1}{2}(h_L u_L + h_R u_R) - \frac{1}{2} \alpha (h_R - h_L), \quad \alpha = \max(|u_L| + c_L, |u_R| + c_R), \quad (37)$$

where $c = \sqrt{g h}$ is the gravity-wave speed (set to zero for pure transport without wave-speed dissipation). The net flux divergence for cell i is

$$-(\nabla \cdot \mathbf{F})_i = -\frac{1}{A_i} \left[(F_x h_y)_{i+1/2} - (F_x h_y)_{i-1/2} + (G_y h_x)_{j+1/2} - (G_y h_x)_{j-1/2} \right], \quad (38)$$

where h_x, h_y are the edge metric factors.

Well-balanced scheme

When topography h_s is present, the scheme reconstructs the free surface $\eta = h + h_s$ instead of h . The fluid depth at each edge is recovered as $h_L = \max(\eta_L - h_{s,\text{edge}}, 0)$, ensuring zero spurious flux for a lake-at-rest state (Audusse et al., 2004).

FC-Gram Spectral Operators

The Fourier Continuation (FC-Gram) method extends periodic Fourier spectral methods to non-periodic domains by appending a smooth continuation interval. On each cubed-sphere face, the 1D function $f(x)$ on $[0, 1]$ is extended to a periodic function on $[0, 1 + d]$ via a Gram polynomial basis in the continuation region.

Extension operator

Given f sampled at n equispaced points, the extension f_{ext} is constructed as

$$f_{\text{ext}}(x) = \sum_{k=0}^{K-1} \alpha_k G_k(x), \quad x \in [1, 1 + d], \quad (39)$$

where G_k are orthogonal Gram polynomials and the coefficients α_k are chosen so that f_{ext} smoothly matches f at both endpoints, yielding a globally smooth periodic extension.

Spectral differentiation

On the extended periodic domain of length $L_{\text{ext}} = 1 + d$, differentiation is performed via FFT:

$$f'_j = \text{IFFT} \left[i k \text{FFT}[f_{\text{ext}}] \right] \Big|_{x_j \in [0,1]}, \quad (40)$$

where k are the Fourier wavenumbers on the extended domain. This achieves spectral accuracy in the interior with controlled Gibbs artifacts at face boundaries.

C-grid-style divergence damping

The FC-Gram variants with divergence damping add 2nd and 4th order selective dissipation of divergent modes:

$$D_{\text{div}} = -\nu_2 \nabla(\nabla \cdot \mathbf{v}) - \nu_4 \nabla(\nabla^2(\nabla \cdot \mathbf{v})), \quad (41)$$

applied to the momentum equations. This selectively damps gravity-wave modes while preserving rotational (balanced) flow, mirroring the divergence damping strategy used in FV3 on its C-D grid.

Atmospheric Dynamics

Shallow Water Equations

The shallow water system is the canonical proving ground for horizontal dynamical cores: it retains the rotational and gravity-wave dynamics of the full equations while collapsing the vertical structure to one layer, so the standard test set of Williamson et al. (1992) can isolate grid-imprinting, dispersion, and conservation errors. In **vector-invariant form** on the cubed sphere:

$$\frac{\partial h}{\partial t} = -\nabla \cdot (h \mathbf{v}), \quad (42)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} + D_u, \quad (43)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} + D_v, \quad (44)$$

where h is the fluid depth, $\mathbf{v} = (u, v)$ the horizontal velocity, f the Coriolis parameter, h_s the bottom topography, the Bernoulli function is $B = \frac{1}{2}(u^2 + v^2) + g(h + h_s)$, ζ is the relative vorticity, and D_u, D_v represent hyperdiffusion. Equation (42) expresses exact mass (volume) conservation in flux form; the momentum equations express the balance between the absolute-vorticity $(\zeta + f)$ flux, the Bernoulli gradient that combines kinetic energy and free-surface pressure, and scale-selective dissipation.

The mass equation (42) is discretised using the finite-volume Rusanov flux (Section 5) with a configurable reconstruction method: MUSCL (2nd order), PPM (4th order), or WENO5 (5th order), selected via `ShallowWaterConfig.fv_limiter`. Momentum uses centred differences with velocity-only biharmonic hyperdiffusion.

Hydrostatic Primitive Equations (σ -coordinates)

Prognostic variables: u, v (winds), T (temperature), p_s (surface pressure). The vertical discretisation uses the energy- and angular-momentum-conserving finite differences of Simmons and Burridge (1981).

$$\frac{\partial p_s}{\partial t} = - \int_0^1 \nabla \cdot (p_s \mathbf{v}) d\sigma, \quad (45)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} - R_d T \frac{\partial \ln p_s}{\partial x} + D_u, \quad (46)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} - R_d T \frac{\partial \ln p_s}{\partial y} + D_v, \quad (47)$$

$$\frac{\partial T}{\partial t} = -\nabla \cdot (T \mathbf{v}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \frac{\partial T}{\partial \sigma} + \kappa T \frac{\omega}{p} + Q, \quad (48)$$

where $B = K + \Phi$, $K = \frac{1}{2}(u^2 + v^2)$, Φ is the geopotential, $\dot{\sigma}$ the vertical velocity in σ -space, $\omega = dp/dt$ the pressure velocity, Q the diabatic heating from physics, and $\kappa = R_d/c_p$. The surface-pressure tendency (45) is the vertically integrated mass continuity that closes the hydrostatic system: with vertical momentum replaced by hydrostatic balance, it is column mass divergence—not vertical acceleration—that sets p_s . The $R_d T \nabla \ln p_s$ term in the momentum equations accounts for the tilt of the σ surfaces, and the adiabatic term $\kappa T \omega/p$ in (48) represents compression warming during descent.

Spectral Primitive Equations (Vorticity–Divergence)

On the Gaussian grid, the momentum equations are recast in the vorticity–divergence form (Bourke, 1972), in which the prognostic scalars $\hat{\zeta}$ and $\hat{\delta}$ replace the vector momentum and the horizontal Laplacian is diagonal in spectral space:

$$\frac{\partial \hat{\zeta}}{\partial t} = \nabla \times [\widehat{(\zeta + f)} \mathbf{v}] + \widehat{\text{vert. adv.}} - \nu \nabla^{2p} \hat{\zeta}, \quad (49)$$

$$\frac{\partial \hat{\delta}}{\partial t} = \nabla \cdot [\widehat{(\zeta + f)} \mathbf{v}] - \nabla^2 (K + \Phi + \widehat{R_d T \ln p_s}) + \widehat{\text{vert. adv.}} - \nu \nabla^{2p} \hat{\delta}, \quad (50)$$

$$\frac{\partial \hat{T}}{\partial t} = -\nabla \cdot (\widehat{T \mathbf{v}}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \widehat{\partial T / \partial \sigma} + \widehat{\kappa T \omega / p}, \quad (51)$$

$$\frac{\partial \widehat{\ln p_s}}{\partial t} = - \int_0^1 \hat{\delta} \, d\sigma. \quad (52)$$

The spectral curl and divergence of flux vectors $(U, V) = (F_u \cos \phi, F_v \cos \phi)$ are

$$\widehat{\text{curl}} = \frac{im}{a} \mathcal{S}[V / \cos^2 \phi] + \frac{1}{a} \mathcal{S}_\mu[U], \quad \widehat{\text{div}} = \frac{im}{a} \mathcal{S}[U / \cos^2 \phi] - \frac{1}{a} \mathcal{S}_\mu[V], \quad (53)$$

where $\mathcal{S}$ and $\mathcal{S}_\mu$ denote the SH analysis with standard and $dP_n^m/d\mu$ Legendre matrices.

Non-Hydrostatic Compressible Euler Equations

In the z^* terrain-following coordinate, the prognostic variables are u, v, w, θ', ρ' (perturbations from a reference state ρ_0, θ_0, π_0). Retaining the full vertical momentum equation (56) admits acoustic and non-hydrostatic gravity-wave modes, which are advanced on a fast time level by the split-explicit integrator below (Skamarock and Klemp, 2008).

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - c_p \theta \frac{\partial \pi'}{\partial x} + D_u - r u, \quad (54)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - c_p \theta \frac{\partial \pi'}{\partial y} + D_v - r v, \quad (55)$$

$$\frac{\partial w}{\partial t} = -\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} - r w, \quad (56)$$

$$\frac{\partial \theta'}{\partial t} = -\nabla \cdot (\theta \mathbf{v}_h) - \frac{w}{J} \frac{\partial \theta}{\partial z^*} + D_\theta, \quad (57)$$

$$\frac{\partial \rho'}{\partial t} = -\frac{1}{J} \left[\nabla_h \cdot (J \rho \mathbf{v}_h) + \frac{\partial(\rho w)}{\partial z^*} \right], \quad (58)$$

where $r(z)$ is the sponge-layer Rayleigh damping coefficient and the Exner function perturbation is

$$\pi' = \pi_0 \left[\left(1 + \frac{\rho'}{\rho_0} \right) \left(1 + \frac{\theta'}{\theta_0} \right) \right]^{R_d/c_v} - \pi_0. \quad (59)$$

Split-explicit time integration

The equations are integrated with an outer SSP-RK3 scheme for the slow tendencies (advection, Coriolis, horizontal pressure gradient, diffusion) and inner forward–backward acoustic substeps for the fast variables (w, θ', ρ') .

Forward step (w).

$$w^{n+1} = w^n + \Delta t_s \left[-\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} \right]. \quad (60)$$

Backward step (ρ', θ').

$$\rho'^{n+1} = \rho'^n - \Delta t_s \frac{\partial(\rho w^{n+1})}{\partial z^*}, \quad \theta'^{n+1} = \theta'^n - \Delta t_s \frac{w^{n+1}}{J} \frac{\partial \theta}{\partial z^*}. \quad (61)$$

An optional **semi-implicit acoustic** solve replaces the forward Euler for w with a tridiagonal system to remove the vertical acoustic CFL constraint.

Acoustic off-centering (v3.7). To damp vertically propagating acoustic/gravity wave noise in long integrations, the density update can be off-centred (Skamarock and Klemp, 2008):

$$\rho_\beta'^{n+1} = (1 + \beta) \rho'^{n+1} - \beta \rho'^n, \quad (62)$$

where $\beta \geq 0$ is the off-centering parameter (**acoustic_off_centering** in the config). For $\beta = 0$ the scheme is centred (neutral); $\beta = 0.1$ is recommended for AMIP/CMIP runs. This adds a small amount of temporal diffusion to the fast modes without tightening the horizontal CFL constraint.

Sponge layer

$$r(z) = r_{\max} \sin^2\left(\frac{\pi}{2} \frac{\max(0, z - z_{\text{bot}})}{z_{\text{top}} - z_{\text{bot}}}\right). \quad (63)$$

Turbulence closure: CRM, LES, and DNS

The diffusion terms D_u, D_v, D_w, D_θ are evaluated in flux form, $D_\phi = \rho_0^{-1} \partial_j(\rho_0 K_\phi \partial_j \phi)$, over the horizontal directions and (optionally) a mass-weighted vertical leg. The same discrete operator spans three turbulence-resolving regimes that differ only in the viscosity K_m and the grid spacing Δx (heat and scalars use $K_h = K_m/\text{Pr}$), selected by the **turbulence_closure** parameter:

- **CRM** (convection-permitting, $\Delta x \sim 1\text{--}4\text{ km}$; **smagorinsky**): the 3-D Smagorinsky–Lilly eddy viscosity (Smagorinsky, 1963) $K_m = (C_s \Delta)^2 \sqrt{\max(0, |S|^2 - \text{Pr} N^2)}$ with $\Delta = (\Delta x \Delta y \Delta z)^{1/3}$, the full strain $|S|^2 = 2S_{ij}S_{ij}$, $C_s \approx 0.19$, and the SAM **dosmagor** moist stratification cutoff N^2 . The energy-containing eddies are entirely sub-grid; this is the regime validated against gSAM for the GATE, LBA, and RCE cases.
- **LES** (large-eddy simulation, $\Delta x \sim 10\text{--}100\text{ m}$; **smagorinsky**): the same closure with $C_s \approx 0.15$ and an optional near-wall limiter $\ell_m = \min(C_s \Delta, \kappa z)$; the energy-containing eddies are resolved. gSAM’s GATE and LBA references are themselves 100 m LES, so this mode uses the identical closure at the identical resolution.
- **DNS** (direct numerical simulation, $\Delta x \rightarrow$ the Kolmogorov scale; **molecular**): no turbulence model; K_m is the constant molecular kinematic viscosity ν with $K_h = \nu/\text{Pr}$ ($\text{Pr} \approx 0.71$ for air), giving the molecular dissipation $\nu \nabla^2 \mathbf{u}$ directly.

Because the three regimes share one discrete flux-form operator, LES and DNS are configuration changes rather than separate code paths; the molecular closure reproduces the analytical viscous decay $\partial_t \hat{u} = -\nu k^2 \hat{u}$ of an isolated shear mode to within the time-integration error.

Atmospheric Physics

Gray Radiation (Frierson 2006)

The gray-radiation scheme (Frierson et al., 2006) replaces the spectral detail of the longwave with a single broadband optical depth that is a prescribed function of pressure and latitude plus a water-vapour contribution, giving an inexpensive but physically self-consistent radiative driving for aquaplanet and idealized-climate experiments.

Longwave optical depth

$$\tau(\sigma, \phi) = [f_l \sigma + (1 - f_l) \sigma^4] [\tau_e + (\tau_p - \tau_e) \sin^2 \phi] + \sum_k \tau_{\text{moist}} q_{v,k} \frac{\Delta p_k}{g}, \quad (64)$$

where the moisture optical depth per layer is $\Delta \tau_{\text{moist},k} = \tau_{\text{moist}} q_{v,k} \Delta p_k / g$, with $\tau_e = 7.2$, $\tau_p = 1.8$, $f_l = 0.2$, $\tau_{\text{moist}} = 0.0115 \text{ m}^2 \text{ kg}^{-1}$.

Two-stream LW fluxes

Layer transmittance and emissivity:

$$\mathcal{T}_k = e^{-D \Delta \tau_k}, \quad \varepsilon_k = 1 - \mathcal{T}_k, \quad D = 1.66. \quad (65)$$

Upward sweep from surface ($F_{\text{sfc}}^\uparrow = \varepsilon_s \sigma_{\text{SB}} T_s^4$):

$$F_k^\uparrow = \mathcal{T}_k F_{k+1}^\uparrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (66)$$

Downward sweep from TOA ($F_0^\downarrow = 0$):

$$F_k^\downarrow = \mathcal{T}_k F_{k-1}^\downarrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (67)$$

Shortwave (Beer–Lambert)

$$F_{\text{SW}}^\downarrow(k) = \frac{S_0}{4} e^{-\tau_{\text{SW},0} \sigma_k}, \quad F_{\text{SW}}^\uparrow(k) = \alpha_s F_{\text{SW}}^\downarrow(\text{sfc}), \quad (68)$$

with $\tau_{\text{SW},0} = 0.22$, $S_0 = 1360 \text{ W m}^{-2}$, $\alpha_s = 0.31$. This follows the Frierson/Isca convention used in the code: only the downward SW beam is absorbed by the atmosphere, while reflected upward SW escapes directly to space.

Heating rate

Defining $F_{\text{net}}^\uparrow = F^\uparrow - F^\downarrow$ (upward positive) at each interface, the heating rate for layer k is

$$\left. \frac{dT}{dt} \right|_{\text{rad},k} = \frac{g}{c_p} \frac{F_{\text{net},k+1/2}^\uparrow - F_{\text{net},k-1/2}^\uparrow}{p_{k+1/2} - p_{k-1/2}}, \quad (69)$$

where $k + \frac{1}{2}$ is the bottom interface and $k - \frac{1}{2}$ the top (pressure increasing downward). Positive divergence of upward flux warms the layer.

RRTMGP Correlated- k Radiation

The RRTMGP backend (Mlawer et al., 1997; Pincus et al., 2019) calls JAX-RRTMGP with correlated- k gaseous optics (128 LW and 112 SW g -points), a two-stream RTE solver, and optional cloud optics. The correlated- k method reorders the line-by-line absorption spectrum by opacity so a handful of quadrature points reproduce the broadband fluxes at a fraction of the cost. Default trace-gas concentrations: $\text{CO}_2=415 \text{ ppmv}$, $\text{CH}_4=1900 \text{ ppbv}$, $\text{N}_2\text{O}=332 \text{ ppbv}$.

SBM Convection (Frierson 2007)

The simplified Betts–Miller (SBM) scheme (Frierson, 2007) relaxes the column toward a moist-adiabatic reference profile whenever it is convectively unstable, conserving column moist enthalpy. Symbols below: Γ_m is the moist-adiabatic lapse rate, q_s the saturation mixing ratio, CAPE the convective available potential energy, $\mathcal{F}$ a smooth trigger, and τ_c the relaxation timescale.

Moist adiabatic lapse rate

$$\Gamma_m = \frac{R_d T}{c_p p} \frac{1 + L_v q_s / (R_d T)}{1 + L_v^2 q_s / (c_p R_v T^2)}, \quad (70)$$

integrated upward via a trapezoidal predictor–corrector.

Saturation mixing ratio (Tetens)

$$e_s = 611.2 \exp\left(\frac{17.67 T_c}{T_c + 243.5}\right), \quad q_s = \frac{\varepsilon e_s}{p - e_s}, \quad \varepsilon = R_d / R_v \approx 0.622. \quad (71)$$

CAPE and trigger

$$\text{CAPE} = R_d \sum_k \max(0, T_{\text{parcel},k} - T_{\text{env},k}) \frac{\Delta p_k}{p_k}. \quad (72)$$

A smooth sigmoid trigger activates convection:

$$\mathcal{F} = \sigma(\kappa_s (\text{CAPE} - \text{CAPE}_c)), \quad \kappa_s = 0.01 \text{ (J/kg)}^{-1}. \quad (73)$$

Enthalpy-conserving relaxation

The reference profile T_{ref} is obtained by adjusting the moist adiabat so that column-integrated moist enthalpy is conserved:

$$\sum_k [c_p (T_{\text{ref},k} - T_k) + L_v (q_{\text{ref},k} - q_{v,k})] \Delta p_k = 0. \quad (74)$$

Newton iteration (2 steps) solves for the temperature offset. Tendencies:

$$\frac{dT}{dt} = \mathcal{F} \frac{T_{\text{ref}} - T}{\tau_c}, \quad \frac{dq_v}{dt} = \mathcal{F} \frac{q_{\text{ref}} - q_v}{\tau_c}, \quad \tau_c = 7200 \text{ s}. \quad (75)$$

Precipitation

$$P = -\frac{1}{g} \sum_k \frac{dq_v}{dt} \Delta p_k. \quad (76)$$

Held–Suarez Forcing

The Held and Suarez (1994) benchmark replaces the full physics with Newtonian relaxation of temperature toward a prescribed radiative equilibrium and Rayleigh friction in the boundary layer, providing a reproducible dynamical-core intercomparison that bypasses parameterisation uncertainty.

Equilibrium temperature

$$T_{\text{eq}} = \max\left(200, [315 - \Delta T_y \sin^2 \phi - \Delta \theta_z \ln(p/p_0) \cos^2 \phi] (p/p_0)^\kappa\right), \quad (77)$$

with $\Delta T_y = 60 \text{ K}$, $\Delta \theta_z = 10 \text{ K}$.

Newtonian relaxation

$$\frac{dT}{dt} = -k_T(\sigma, \phi) (T - T_{\text{eq}}), \quad k_T = k_a + (k_s - k_a) \max\left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b}\right) \cos^4 \phi, \quad (78)$$

with $k_a = 1/(40 \text{ day})$, $k_s = 1/(4 \text{ day})$, $\sigma_b = 0.7$.

Rayleigh friction

$$\frac{d\mathbf{v}}{dt} = -k_v(\sigma) \mathbf{v}, \quad k_v = k_f \max\left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b}\right), \quad k_f = 1/(1 \text{ day}). \quad (79)$$

Louis (1979) Turbulence

The Louis (1979) scheme parameterises vertical eddy fluxes through a mixing length and Richardson-number-dependent stability functions, giving a smooth, differentiable closure for the boundary-layer diffusivities K_m and K_h .

Mixing length

$$l = \frac{\kappa_{vK} z}{1 + \kappa_{vK} z/l_{\max}}, \quad \kappa_{vK} = 0.4, \quad l_{\max} = 100 \text{ m}. \quad (80)$$

Richardson number

$$\text{Ri} = \frac{N^2}{S^2}, \quad N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}, \quad S^2 = \left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2. \quad (81)$$

Stability functions

$$f_m(\text{Ri}) = \begin{cases} 1 - \frac{2b \text{ Ri}}{1 + 3bc l^2 |\text{Ri}|^{1/2} / \Delta z^2} & \text{Ri} < 0, \\ \frac{1}{1 + 2b \text{ Ri} / \sqrt{1 + d \text{ Ri}}} & \text{Ri} \geq 0, \end{cases} \quad (82)$$

with $b = c = d = 5$. Eddy diffusivities: $K_m = l^2 S f_m$, $K_h = K_m$.

Vertical Diffusion (Implicit Thomas Algorithm)

The backward-Euler system $-a_k \varphi_{k-1} + b_k \varphi_k - c_k \varphi_{k+1} = d_k$ is solved with the Thomas algorithm.

Tridiagonal coefficients.

$$a_k = \frac{\Delta t K_{k-1/2}}{\rho_k \Delta z_k \Delta z_{k-1/2}}, \quad c_k = \frac{\Delta t K_{k+1/2}}{\rho_k \Delta z_k \Delta z_{k+1/2}}, \quad b_k = 1 + a_k + c_k. \quad (83)$$

Forward sweep.

$$w_k = \frac{a_k}{b'_{k-1}}, \quad b'_k = b_k - w_k c_{k-1}, \quad d'_k = d_k + w_k d'_{k-1}. \quad (84)$$

Back substitution.

$$\varphi_k = \frac{d'_k + c_k \varphi_{k+1}}{b'_k}. \quad (85)$$

Surface Fluxes

Three bulk flux schemes are available, selected per tile.

Constant coefficients (default)

$$\tau_x = -\rho C_D |\mathbf{V}| u, \quad \tau_y = -\rho C_D |\mathbf{V}| v, \quad (86)$$

$$\text{SH} = \rho c_p C_H |\mathbf{V}| (T_s - T_a), \quad (87)$$

$$\text{LH} = \rho L_v C_E |\mathbf{V}| (q_s - q_a), \quad (88)$$

with $C_D = C_H = C_E = 1.5 \times 10^{-3}$ and $|\mathbf{V}| = \sqrt{u^2 + v^2 + 10^{-4}}$.

Monin–Obukhov Similarity Theory (MOST)

Stability-dependent transfer coefficients via iterative Obukhov-length computation using the Businger–Dyer flux–profile relationships (Businger et al., 1971).

Obukhov length.

$$L = -\frac{u_*^3 T_v}{\kappa_{\text{vK}} g (\theta_* + 0.61 T_a q_*)}, \quad (89)$$

where u_* , θ_* , q_* are friction velocity, temperature scale, and moisture scale.

Stability parameter. $\zeta = z_{\text{ref}}/L$.

Businger–Dyer stability functions.

$$\psi_m(\zeta) = \begin{cases} 2 \ln\left(\frac{1+x}{2}\right) + \ln\left(\frac{1+x^2}{2}\right) - 2 \arctan x + \frac{\pi}{2}, & \zeta < 0, \quad x = (1 - 16\zeta)^{1/4}, \\ -5\zeta, & \zeta \geq 0, \end{cases} \quad (90)$$

$$\psi_h(\zeta) = \begin{cases} 2 \ln\left(\frac{1+y}{2}\right), & \zeta < 0, \quad y = (1 - 16\zeta)^{1/2}, \\ -5\zeta, & \zeta \geq 0. \end{cases} \quad (91)$$

Transfer coefficients.

$$C_D = \left[\frac{\kappa_{\text{vK}}}{\ln(z/z_0) - \psi_m(\zeta)} \right]^2, \quad C_H = \frac{\kappa_{\text{vK}}^2}{[\ln(z/z_0) - \psi_m(\zeta)][\ln(z/z_{0t}) - \psi_h(\zeta)]}. \quad (92)$$

Iterative solver. The system u_* , θ_* , q_* , z_0 , z_{0t} , z_{0q} , L is solved iteratively (default 5 iterations) using `jax.lax.fori_loop` for full JAX differentiability.

COARE 3.0 roughness (Fairall et al. 2003)

Over ocean, the COARE 3.0 bulk algorithm (Fairall et al., 2003) updates the roughness lengths each iteration from the evolving friction velocity:

$$z_0 = \alpha_c \frac{u_*^2}{g} + 0.11 \frac{\nu}{u_*}, \quad z_{0t} = \min(1.15 \times 10^{-4}, 5.5 \times 10^{-5} \text{Re}_*^{-0.6}), \quad (93)$$

where $\alpha_c = 0.011$ is the Charnock parameter and $\text{Re}_* = u_* z_0 / \nu$.

Large & Yeager (2004) roughness

The Large and Yeager (2004) bulk formulae specify an empirical neutral drag coefficient as a function of the neutral 10-m wind speed U_{10N} :

$$C_{DN} \times 10^3 = \frac{2.7}{U_{10N}} + 0.142 + 0.0764 U_{10N}, \quad (94)$$

with roughness recovered as $z_0 = 10 / \exp(\kappa_{vK} / \sqrt{C_{DN}})$. Stanton and Dalton numbers follow the stability-corrected LY04 parameterisation.

Additional Turbulence Schemes

Smagorinsky

Constant eddy diffusivity baseline: $K_m = K_h = \text{const.}$

YSU (Yonsei University)

Nonlocal K-profile boundary-layer scheme with explicit entrainment treatment at the PBL top.

CLUBB-lite

Higher-order prognostic moment closure with five prognostic variables ($\overline{w'^2}$, $\overline{\theta_l'^2}$, $\overline{r_t'^2}$, $\overline{w'\theta_l'}$, $\overline{w'r_t'}$) and cloud fraction diagnosis via assumed bivariate Gaussian PDF.

EDMF Turbulence

Eddy-diffusivity mass-flux unified PBL framework combining diffusive and convective transport in a single scheme.

Additional Convection Schemes

Kuo (1974)

Column moisture-excess scheme (Kuo, 1974) partitioning excess moisture between heating (fraction α_{heat}) and moistening ($1 - \alpha_{\text{heat}}$).

Mass-Flux Convection (Arakawa–Wu Type)

Entraining-detraining plume model with bulk mass flux, CAPE closure, and configurable entrainment/detrainment rates.

EDMF Convection

Combined eddy-diffusivity and mass-flux framework for shallow and deep convection.

Deep Convective Adjustment (DCA)

Bottom-to-top adjustment with smooth sigmoid triggers for JAX differentiability.

Gravity Wave Drag

Six GWD schemes are available:

- **Rayleigh friction:** Simple linear damping above a threshold height.
- **Lindzen (1981):** Orographic GWD with critical-level absorption (Lindzen, 1981).
- **McFarlane (1987):** Enhanced orographic scheme with improved saturation (McFarlane, 1987).
- **Hines (1997):** Doppler-spread parameterisation for non-orographic GWD (Hines, 1997).
- **Prognostic Spectral:** Multi-azimuthal, multi-wavenumber spectrum with prognostic wave flux and vertical propagation via `jax.lax.scan`.
- **ML Emulator:** Equinox MLP surrogate for learned GWD.

Cloud Microphysics

Five microphysics schemes:

- **Kessler (1969):** Warm-rain autoconversion with threshold cloud water (Kessler, 1969).
- **Seifert–Beheng (2001):** Two-moment warm rain with prognostic number concentration (Seifert and Beheng, 2001).
- **Morrison (2005):** Double-moment ice + liquid scheme (Morrison et al., 2005).
- **Thompson:** Hybrid-moment scheme with ice crystal habits (Thompson et al., 2008).
- **Sundqvist (1988) and Xu–Randall (1996)** cloud fraction diagnostics (Sundqvist et al., 1989; Xu and Randall, 1996).

`cloud_fraction.compute_cloud_properties` dispatches on `config.scheme` and accepts only "sundqvist" or "xu_randall"; any other literal raises `ValueError` (no silent fallback). To disable cloud diagnostics, set `cloud_scheme="none"` on the caller's config so that `compute_cloud_properties` is skipped entirely upstream; "none" is not itself a valid scheme inside the dispatch.

Ocean Dynamics

Boussinesq Hydrostatic Primitive Equations

The ocean solves the hydrostatic Boussinesq primitive equations in the finite-volume, split-explicit free-surface formulation of Marshall et al. (1997), in which density variations are retained only in the buoyancy term and the fast external (barotropic) mode is integrated on a short time level. In vector-invariant form on the cubed sphere:

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - \frac{1}{\rho_0} \frac{\partial p'}{\partial x} + A_h \nabla^2 u + \frac{\partial}{\partial z} \left(A_v \frac{\partial u}{\partial z} \right) - w \frac{\partial u}{\partial z}, \quad (95)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - \frac{1}{\rho_0} \frac{\partial p'}{\partial y} + A_h \nabla^2 v + \frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right) - w \frac{\partial v}{\partial z}. \quad (96)$$

The Coriolis force is split: depth-mean $f\mathbf{v}$ in the barotropic substeps, baroclinic shear $f\mathbf{v}'$ in the slow tendencies.

Tracer equations (skew-symmetric)

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \left[-\mathbf{v} \cdot \nabla \phi - \nabla \cdot (\phi \mathbf{v}) \right] - w \frac{\partial \phi}{\partial z} + K_h \nabla^2 \phi + \frac{\partial}{\partial z} \left(K_v \frac{\partial \phi}{\partial z} \right). \quad (97)$$

Free-surface equation

$$\frac{\partial \eta}{\partial t} = - \sum_k \nabla \cdot (h_k \mathbf{v}_k). \quad (98)$$

Barotropic Solver (Forward–Backward, Semi-Implicit Coriolis)

The barotropic system is solved with split-explicit forward–backward substeps:

$$\frac{\partial \eta}{\partial t} = -\nabla \cdot (H \bar{\mathbf{V}}), \quad \frac{\partial \bar{\mathbf{V}}}{\partial t} = f \hat{\mathbf{k}} \times \bar{\mathbf{V}} - g \nabla \eta + \mathbf{F}_{\text{slow}}. \quad (99)$$

Semi-implicit Coriolis. With $\alpha = \frac{1}{2} f \Delta t_s$:

$$\bar{U}^{n+1} = \frac{R_u + \alpha R_v}{1 + \alpha^2}, \quad \bar{V}^{n+1} = \frac{R_v - \alpha R_u}{1 + \alpha^2}, \quad (100)$$

where $R_u = \bar{U}^n + \alpha \bar{V}^n - \Delta t_s g \partial \eta / \partial x$ and $R_v = \bar{V}^n - \alpha \bar{U}^n - \Delta t_s g \partial \eta / \partial y$.

Equation of State (Wright 1997)

The ocean prognostic temperature variable T is **potential temperature** [degC]. The Wright (1997) equation of state computes in-situ density directly from potential temperature θ , salinity S [PSU], and pressure p [Pa]:

$$\rho = \frac{p + p_0(\theta, S)}{\lambda(\theta, S) + \alpha_0(\theta, S)(p + p_0)}, \quad (101)$$

where

$$\alpha_0 = a_0 + a_1 \theta + a_2 S, \quad (102)$$

$$p_0 = (b_0 + b_4 S) + \theta(b_1 + \theta(b_2 + b_3 \theta) + b_5 S), \quad (103)$$

$$\lambda = (c_0 + c_4 S) + \theta(c_1 + \theta(c_2 + c_3 \theta) + c_5 S). \quad (104)$$

Coefficients follow the MOM6 implementation. The density perturbation is $\rho' = \rho(\theta, S, p) - \rho_0$.

Linear Equation of State

A configurable linear EOS is also available for idealised experiments:

$$\rho = \rho_0 [1 - \alpha_T (T - T_{\text{ref}}) + \beta_S (S - S_{\text{ref}})], \quad (105)$$

with typical values $\alpha_T = 2 \times 10^{-4} \text{ K}^{-1}$, $\beta_S = 7.4 \times 10^{-4} \text{ PSU}^{-1}$. The linear EOS is pressure-independent (no hydrostatic pressure coupling iterations required), enabling faster compilation and execution. It is selected via `eos="linear"` in the ocean configuration, with parameters set via a `LinearEOSConfig` NamedTuple. A dispatcher `make_eos_fn()` returns the appropriate EOS callable.

Hydrostatic Pressure and Vertical Velocity

Baroclinic pressure by downward integration:

$$p'_k = \rho_0 g \eta + \sum_{k'=0}^{k-1} \rho'_{k'} g \Delta z_{k'} + \frac{1}{2} \rho'_k g \Delta z_k. \quad (106)$$

Vertical velocity diagnosed bottom-up from $w(\text{floor}) = 0$:

$$w_{k-1/2} = - \sum_{k'=k}^{N-1} \nabla \cdot (h_{k'} \mathbf{v}_{k'}). \quad (107)$$

Single-level barotropic mode ($N_{\text{lev}} = 1$). The z-star coordinate supports $N_{\text{lev}} = 1$, reducing exactly to the nonlinear shallow-water equations: $h = H + \eta$, with no baroclinic modes. Vertical mixing operators return zero for $N_{\text{lev}} < 2$. This enables MITgcm-style barotropic gyre experiments with a single depth-integrated layer.

Ocean Vertical Mixing

Three vertical mixing schemes are available:

K-Profile Parameterisation (KPP, Large et al. 1994)

The K-profile parameterisation (Large et al., 1994) diagnoses the boundary-layer depth h_{bl} from the bulk Richardson number criterion $\text{Ri}_b(h) = \text{Ri}_c = 0.3$. Within the boundary layer, diffusivity follows a cubic shape function:

$$K(z) = h_{\text{bl}} w_* G(\sigma), \quad G(\sigma) = \sigma(1 - \sigma)^2, \quad \sigma = -z/h_{\text{bl}}, \quad (108)$$

where w_* is a turbulent velocity scale incorporating wind and buoyancy forcing. Non-local transport of temperature and salinity: $\gamma_T = -C_s (\overline{w'\theta'})_0 / (w_* h_{\text{bl}})$.

Richardson-Number Mixing

Interior shear-driven diffusivity:

$$K_v = K_{v,0} (1 + 5 \text{ Ri})^{-1}, \quad \text{Ri} = \frac{N^2}{S^2}. \quad (109)$$

Constant Mixing

Uniform K_v (default $10^{-4} \text{ m}^2 \text{ s}^{-1}$) for idealised experiments.

Ocean Lateral Mixing

Harmonic (Laplacian) Diffusion

$F_h = A_h \nabla^2 \phi$ with constant or Smagorinsky-type coefficient.

Biharmonic (∇^4) Diffusion

$F_h = -A_4 \nabla^4 \phi$, providing scale-selective damping of grid-scale noise.

Gent–McWilliams / Redi (GM/Redi)

Mesoscale eddy parameterisation (Gent and McWilliams, 1990) with bolus velocity $\mathbf{v}^* = -\kappa_{\text{GM}} \nabla_h s$ where s is the isopycnal slope; the adiabatic stirring by unresolved eddies is represented as an advective flattening of isopycnals. Isopycnal diffusion rotates the diffusion tensor to align with isopycnal surfaces (Redi, 1982). Default $\kappa_{\text{GM}} = 1000 \text{ m}^2 \text{ s}^{-1}$.

Ocean Bottom Drag

Two formulations:

- **Linear:** $\tau_b = -r \mathbf{u}_b$ with configurable r .
- **Quadratic:** $\tau_b = -C_d |\mathbf{u}_b| \mathbf{u}_b$ with $C_d = 2.5 \times 10^{-3}$.

Ocean Surface Forcing

Three modes, available on cubed-sphere, lat-lon, and MPAS grids:

- **Prescribed:** Fixed wind stress and heat flux profiles (cosine-latitude, single-gyre, double-gyre, constant).
- **Restoring:** SST/SSS restoring toward climatology (Haney-style) with configurable timescale.
- **Bulk formulas:** COARE-like air-sea fluxes with drag, Stanton, and Dalton coefficients.

On MPAS Voronoi meshes, cell-centred wind stress (τ_x, τ_y) is projected onto TRiSK edge normals via $\tau_n = \tau_x \cos \alpha_e + \tau_y \sin \alpha_e$ where α_e is the edge normal angle. An MPAS-specific physics factory (`make_mpas_ocean_physics`) handles this projection.

Shortwave Penetration

Subsurface absorption following Jerlov (1968) water type classification (types I, IA, IB, II, III) with two-band exponential decay:

$$I(z) = I_0 [R e^{-z/\zeta_1} + (1 - R) e^{-z/\zeta_2}], \quad (110)$$

where R, ζ_1, ζ_2 depend on the water type.

Ocean Convection

Enhanced Diffusion

Convective adjustment via enhanced diffusivity K_{conv} where $N^2 < 0$, with smooth transitions.

Plume Model

Entrainment mass-flux ocean convection with entrainment rate $\alpha_{\text{plume}} = 0.1$ and background detrainment $\varepsilon = 10^{-3} \text{ m}^{-1}$.

Ocean Conservation Fixers

Grid-specific volume, heat, and salt conservation fixers for cubed-sphere, lat-lon, and MPAS grids. Uniform additive correction per unit volume after each timestep. All three grids compute corrections simultaneously from the original pre-fix state (avoiding order-dependent bias) and upcast to accumulation precision before global reductions to prevent catastrophic cancellation in float32.

Limitation: These are uniform additive fixers that violate local conservation and act as spurious globally-uniform diapycnal mixing. The proper solution is flux-form tracer advection with barotropic-baroclinic flux reconciliation (Hallberg 1997, Higdon 2005).

Tracer Advection Schemes

Tracer transport in `src/legoesm/ocean/advection.py` and `advection_som.py` provides upwind, TVD (Van Leer / Sweby), DST3, PPM, PPM-FCT, WENO5/7, and Second-Order Moments (SOM) schemes (Prather, 1986). The volume-weighted variance $\text{Var}(T)$ on the Eady uniform channel ranks them: `som + Smagorinsky` ($C_{\text{Smag}}=0.2$) preserves 97.1% of $\text{Var}(T)$ at 600 d in the weak-forcing regime; `tvd` preserves 94% at $\leq 2\%$ runtime overhead vs. `upwind` (recommended production default); `ppm_fct` is gated on the sign-split Zalesak/Kuzmin limiter (open issue #9). Forcing regime (strong $U=0.8 \text{ m s}^{-1}$ vs. weak $U=0.2$) dominates absolute mixing; the weak-forcing setup is the scientifically defensible ocean-interior benchmark.

Realistic-Geometry Spinup

Global Wolfe–Cessi spinups on real ETOPO/GEBCO bathymetry are supported on the lat-lon C-grid via `src/legoesm/ocean/bathymetry.py` (auto-detect ETOPO/GEBCO/ERDDAP/CF, Laplacian smoothing, MEO r -cap, polar-cap mask, isolated-basin filling, strait enforcement). Stability conditioners: AL81 12-corner-triad PV-flux closure with partial-cell extension (Le Sommer et al., 2009, `pv_flux_al81_partial_cell`); MOM6-style $\cos^4 \phi$ biharmonic-viscosity scaling (Griffies and Hallberg, 2000, `biharmonic_scaling_factor`); SMC03 pressure-gradient force; implicit Crank–Nicolson barotropic solver. Validated 20-yr global-overturning spinup at 1° delivers canonical thermohaline structure (branch `realistic-geometry-full1`, PR #224).

Time Integration

SSP-RK3 (Shu–Osher 1988)

The three-stage strong-stability-preserving Runge–Kutta method (Shu and Osher, 1988), whose convex-combination structure guarantees that the total-variation-diminishing property of the forward-Euler step is inherited at higher order:

$$\mathbf{u}^{(1)} = \mathbf{u}^n + \Delta t F(\mathbf{u}^n), \quad (111)$$

$$\mathbf{u}^{(2)} = \frac{3}{4}\mathbf{u}^n + \frac{1}{4}\mathbf{u}^{(1)} + \frac{1}{4}\Delta t F(\mathbf{u}^{(1)}), \quad (112)$$

$$\mathbf{u}^{n+1} = \frac{1}{3}\mathbf{u}^n + \frac{2}{3}\mathbf{u}^{(2)} + \frac{2}{3}\Delta t F(\mathbf{u}^{(2)}). \quad (113)$$

SSP-RK(4,3) and SSP-RK(5,4) variants are also available.

Robert–Asselin–Williams (RAW) Time Filter (v3.7)

The leapfrog scheme used by the semi-implicit spectral PE solver requires a time filter to damp the computational mode. The Williams (2009) modification of the Robert–Asselin filter splits the correction between the current and next time levels, restoring second-order accuracy:

$$d^n = \frac{\gamma}{2}(X^{n-1} - 2X^n + X^{n+1}), \quad (114)$$

$$X_{\text{filt}}^n = X^n + (1 - \alpha) d^n, \quad (115)$$

$$X_{\text{filt}}^{n+1} = X^{n+1} + \alpha d^n. \quad (116)$$

With $\alpha = 0.5$ (default), this is the RAW filter; $\alpha = 0$ recovers the original Robert–Asselin filter. The original RA filter introduces a first-order phase error that causes slow energy drift over climate-length integrations; the RAW modification eliminates this while preserving the same damping rate for the computational mode.

Adaptive Hyperdiffusion (v3.7)

The maximum stable coefficient for n -th order hyperdiffusion under forward Euler is

$$\nu_n = S \cdot \frac{C(n) \Delta x^n}{\Delta t}, \quad C(2) = \frac{1}{2}, C(4) = \frac{1}{8}, C(6) = \frac{1}{48}, \quad (117)$$

where $S \in (0, 1]$ is a safety factor (default 0.5). The function `adaptive_hyperdiff_coeff(dx_min, dt, order, safety)` eliminates manual tuning of the hyperdiffusion coefficient for each resolution and timestep.

Split-Explicit Methods

Each outer RK stage computes slow tendencies once, updates u, v and tracers, then runs N_s forward-backward acoustic substeps on the fast variables (w, θ', ρ') with substep size $\Delta t_s = \Delta t / N_s$.

Land, Ice, and Coupler

Slab Land Model

Energy balance

$$C_s d_s \frac{dT_s}{dt} = (1 - \alpha_l) SW^\downarrow + \varepsilon_l (LW^\downarrow - \sigma_{SB} T_s^4) - SH - LH. \quad (118)$$

Bucket hydrology

$$\frac{dW}{dt} = P - E, \quad W \in [0, W_{\max}]. \quad (119)$$

Moisture availability $\beta = \beta_{\min} + (1 - \beta_{\min}) W / W_{\max}$ modulates evaporation via $q_{\text{sfc}} = \beta q_s(T_s)$.

Surface humidity consistency. All slab surface tiles (land, sea ice, lake) recompute q_{sfc} from the *updated* surface temperature T_s^{n+1} (and updated W^{n+1} for land) before constructing the `TileResponse`. This ensures that the returned q_{sfc} and T_{surface} are thermodynamically consistent at the end of the timestep.

Sea-Ice Model

Surface temperature

$$\rho_i c_i \frac{h_{\text{eff}}}{2} \frac{dT_i}{dt} = Q_{\text{sfc}} - \frac{k_i(T_f - T_i)}{h_{\text{eff}}}, \quad h_{\text{eff}} = \max(h_i, h_{\min}). \quad (120)$$

Growth and melt

$$\frac{dh_i}{dt} = \frac{F_{\text{cond}} - F_{\text{ocean}}}{\rho_i L_f}. \quad (121)$$

Concentration

$$\frac{dc}{dt} = \begin{cases} (dh/dt)(1 - c)/h_{\text{new}} & \text{growing,} \\ (dh/dt) c / h_{\text{eff}} & \text{melting.} \end{cases} \quad (122)$$

EVP Rheology (Hunke & Dukowicz 1997)

The elastic-viscous-plastic rheology (Hunke and Dukowicz, 1997) adds an artificial elastic term to the viscous-plastic ice stress so that the momentum equation can be subcycled explicitly, avoiding an implicit solve. When `dynamics="evp"`, the ice momentum equation is

$$m \frac{\partial \mathbf{u}_i}{\partial t} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_w - m f \hat{\mathbf{k}} \times \mathbf{u}_i - mg \nabla \eta + \nabla \cdot \boldsymbol{\sigma}, \quad (123)$$

where $\boldsymbol{\tau}_a, \boldsymbol{\tau}_w$ are air and ocean drag, and $\boldsymbol{\sigma}$ is the internal stress tensor from an elliptic yield curve with eccentricity $e = 2$. Ice strength:

$$P = P^* h \exp(-C(1 - c)), \quad P^* = 27.5 \text{ kPa}, \quad C = 20. \quad (124)$$

The elastic term provides subcycling stability ($N_{\text{evp}} = 120$ substeps).

Multi-category ice

For $N_{\text{cat}} > 1$, per-category state (h_k, T_k, c_k) evolves independently under thermodynamics, then the linear remapping of Lipscomb (2001) redistributes ice across thickness categories.

Land Carbon Cycle and Stomatal Conductance

DALEC-990 Carbon Pools

Six prognostic pools: labile C_{lab} , foliage C_f , root C_r , wood C_w , litter C_{lit} , soil organic matter C_{SOM} . GPP via light-use efficiency; decomposition via Q_{10} temperature sensitivity with moisture limitation.

Farquhar (1980) Photosynthesis

The Farquhar et al. (1980) biochemical model expresses the leaf assimilation rate as the smooth minimum of a Rubisco-limited rate W_c and an electron-transport-limited rate W_j , less leaf respiration R_d :

$$A = \min(W_c, W_j) - R_d, \quad W_c = V_{c,\max} \frac{C_i - \Gamma^*}{C_i + K_c(1 + O/K_o)}, \quad W_j = \frac{J(C_i - \Gamma^*)}{4(C_i + 2\Gamma^*)}, \quad (125)$$

where the minimum is a smooth (differentiable) function. $V_{c,\max}$ and $J_{\max}$ follow peaked Arrhenius temperature responses (Bernacchi et al., 2001). Soil moisture stress: $V_{c,\max} \leftarrow V_{c,\max} \beta_{\text{soil}}$.

Stomatal conductance

Ball–Berry (Ball et al., 1987): $g_s = g_0 + g_1 A_{\text{RH}}/C_s$.

Medlyn (Medlyn et al., 2011): $g_s = g_0 + 1.6(1 + g_1/\sqrt{\text{VPD}}) A/C_s$.

Jarvis (Jarvis, 1976): $g_s = g_{s,\max} f(\text{PAR}) f(T) f(\text{VPD}) f(\theta)$ (CO₂-independent; used when carbon cycle is inactive).

The coupled A – g_s – C_i system is solved by 5 fixed-point iterations (unrolled for JAX JIT).

Plant Functional Types (PFTs)

CLM5-style weighted-average lookup over 17 Plant Functional Types (needleleaf/broadleaf trees, shrubs, grasses, crops), with 12 trainable land parameters per PFT: vegetation albedo, emissivity, roughness length, maximum soil water, soil heat capacity, soil depth, root depth, wilting point, field capacity, $V_{c,\max,25}$, LCMA, and stomatal slope g_1 . Each parameter is bounded via sigmoid transforms:

$$p = p_{\min} + (p_{\max} - p_{\min}) \sigma(x_{\text{raw}}). \quad (126)$$

Three parameter provider modes:

- **Constant:** Spatially uniform default values.
- **PFT:** Weighted average $\bar{p} = \sum_{k=1}^{17} f_k p_k$ from the PFT lookup table.
- **Neural:** 3-hidden-layer FCNN (GELU, 48 hidden units) mapping static features (lat/lon, soil type, elevation, precipitation, temperature, forest age) to land parameters.

DifferLand Training

End-to-end differentiable land model training with three carbon cycle modes: “none” (disabled), “differland” (prognostic DALEC-990 with trainable allocation and decomposition parameters), and “seasonal” (prescribed LAI). Phenology uses hemisphere-aware Gaussian windows for leaf emergence/fall. Decomposition follows Q_{10} temperature sensitivity:

$$k(T, \theta) = k_0 Q_{10}^{(T-T_{\text{ref}})/10} f_w(\theta). \quad (127)$$

NPP allocation is sequential: foliar $\rightarrow$ labile $\rightarrow$ root $\rightarrow$ wood.

Tile Coupler

Surface fluxes are area-weighted blends over ocean, land, sea-ice, and lake tiles:

$$X_{\text{blend}} = f_{\text{ocn}} X_{\text{ocn}} + f_{\text{land}} X_{\text{land}} + f_{\text{ice}} X_{\text{ice}} + f_{\text{lake}} X_{\text{lake}}. \quad (128)$$

Fluxes are time-averaged over each coupling window before being passed to the atmosphere.

Ocean albedo. The constant ocean albedo α_{ocn} is controlled by `CouplerConfig.ocean_albedo` (default 0.06). When `ocean_albedo_config.method="zenith"`, a Briegleb-type zenith-angle formula is used (Briegleb et al., 1986):

$$\alpha_{\text{ocn}} = \frac{0.026}{1.1 \mu_0^{1.7} + 0.065} + 0.15 (\mu_0 - 0.1)(\mu_0 - 0.5)(\mu_0 - 1.0), \quad (129)$$

where $\mu_0 = \cos \theta_z$ is the cosine of the solar zenith angle.

External forcing. Time-varying greenhouse gases (CO_2 , CH_4 , N_2O), total solar irradiance, and monthly zonal-mean ozone and aerosol climatologies can be read from NetCDF files and linearly interpolated in time via the `ExternalForcingConfig` interface. Ozone and aerosol data are loaded but not yet connected to the radiation solver.

Two-layer lake model. Epilimnion (mixed layer, default $h_{\text{epi}} = 5$ m) and hypolimnion (deep layer, $h_{\text{hypo}} = 20$ m) connected by wind-enhanced vertical mixing:

$$k_{\text{eff}} = k_{\text{mix}} (1 + \alpha_{\text{wind}} |\mathbf{V}|), \quad (130)$$

with surface energy balance driving the epilimnion temperature and mixing flux $F_{\text{mix}} = \rho c k_{\text{eff}} \Delta T / d_{\text{mid}}$ coupling the two layers. Lake freezing is clamped at $T_f = 273.15$ K. Bulk flux options: constant coefficients or iterative MOST.

Machine Learning: Spherical Fourier Neural Operator

The Spherical Fourier Neural Operator (Bonev et al., 2023) generalises the Fourier neural operator (Li et al., 2021) to the sphere by replacing the planar FFT with a spherical-harmonic transform, so the learned global convolution is rotation-equivariant and free of the polar artefacts a lat-lon FFT would introduce; it provides the learned-dynamics backbone in the same spirit as the neural general circulation model of Kochkov et al. (2024). The SFNO (ACE/FourCastNet v2 style) operates on the Gaussian grid.

Architecture

1. **Encoder:** Pointwise linear $n_{\text{in}} \rightarrow n_{\text{emb}}$.
2. N **processor blocks**, each containing:
 - Layer normalisation,
 - SH analysis $\rightarrow$ learned spectral convolution $\rightarrow$ SH synthesis,
 - GELU activation,
 - Pointwise MLP (two layers with expansion factor),
 - Skip connection.
3. **Decoder:** Pointwise linear $n_{\text{emb}} \rightarrow n_{\text{out}}$.
4. **Big-skip residual:** $\text{output} = \text{input} + \text{decoder}(\dots)$.

Spectral convolution

In the SH coefficient space:

$$\hat{f}_{nm}^{\text{out}} = W_{nm} \hat{f}_{nm}^{\text{in}}, \quad (131)$$

where $W_{nm} \in \mathbb{R}^{n_{\text{emb}} \times n_{\text{emb}}}$ is a learnable weight matrix per wavenumber pair.

Neural and Learned Physics

NeuralPhysics

Equinox MLP column model with hybrid mode blending learned corrections with conventional physics via a configurable blending coefficient.

LearnedColumn

Per-column physics replacement following Rasp et al. (2018), mapping the full column state to temperature and moisture tendencies via a neural network, trainable end-to-end with the dynamical core.

Conservation Corrections

Atmosphere

Dry-air mass is corrected by a uniform adjustment to p_s that restores the area-weighted global integral:

$$\Delta \bar{p} = \frac{\sum_i A_i p_{s,i}^{\text{new}} - \sum_i A_i p_{s,i}^{\text{old}}}{\sum_i A_i}, \quad p_{s,i}^{\text{corr}} = p_{s,i}^{\text{new}} - \Delta \bar{p}, \quad (132)$$

so that $\sum_i A_i p_{s,i}^{\text{corr}} = \sum_i A_i p_{s,i}^{\text{old}}$ (global dry mass conserved exactly). Column moisture is rescaled: $q^{\text{corr}} = q^{\text{new}} \cdot Q_{\text{old}}/Q_{\text{new}}$.

Initial-mass anchoring (v3.7). For long AMIP/CMIP integrations, the fixer can anchor to the *initial* mass $M_0 = \sum_i A_i p_{s,i}^{\text{init}}$ rather than the previous step's mass, preventing cumulative $O(\varepsilon)$ drift over millions of timesteps. The correction becomes

$$\Delta \bar{p} = \frac{\sum_i A_i p_{s,i}^{\text{new}} - M_0}{\sum_i A_i}. \quad (133)$$

Set via `model.set_initial_mass(state)` or `anchor_mass_to_initial=True` in the config. All accumulations use float64 regardless of the state precision.

Ocean

Volume conservation: $\eta^{\text{corr}} = \eta^{\text{new}} + (\sum \eta_{\text{old}} A - \sum \eta_{\text{new}} A) / \sum A$.

Heat and salt conservation: uniform additive correction per unit volume.

Physical Constants

Symbol	Name	Value	Unit
g	Gravitational acceleration	9.80616	m s^{-2}
R_d	Dry-air gas constant	287.04	$\text{J kg}^{-1} \text{K}^{-1}$
c_p	Dry-air heat capacity	1005.0	$\text{J kg}^{-1} \text{K}^{-1}$
R_v	Water-vapour gas constant	461.5	$\text{J kg}^{-1} \text{K}^{-1}$
L_v	Latent heat of vapourisation	2.501×10^6	J kg^{-1}
κ	R_d/c_p	0.285	—
p_0	Reference pressure	10^5	Pa
σ_{SB}	Stefan–Boltzmann constant	5.67×10^{-8}	$\text{W m}^{-2} \text{K}^{-4}$
κ_{vK}	von Kármán constant	0.4	—
Ω	Earth rotation rate	7.292×10^{-5}	rad s^{-1}
$R_{\oplus}$	Earth radius	6.371×10^6	m
ρ_0	Ocean reference density	1025	kg m^{-3}

Ocean Biogeochemistry

Two schemes: abiotic (DIC + ALK only) and NPZD ecosystem.

Carbonate Chemistry and Air-Sea CO_2 Exchange

The ocean surface $p\text{CO}_2$ is computed from DIC, ALK, T , S via the carbonate-system solution of Follows et al. (2006). Air-sea flux (Wanninkhof, 2014):

$$F_{\text{CO}_2} = k_w K_0 (p\text{CO}_{2,\text{ocean}} - p\text{CO}_{2,\text{atm}}), \quad (134)$$

where $k_w = 0.251 U_{10}^2 (\text{Sc}/660)^{-1/2}$ is the gas transfer velocity [cm/hr] and K_0 is Henry's law solubility.

NPZD Ecosystem Model

Four biological tracers: NO_3 , phytoplankton P , zooplankton Z , detritus D (Fasham et al., 1990).

$$\frac{\partial P}{\partial t} = \mu_P P - G - m_P P, \quad (135)$$

$$\frac{\partial Z}{\partial t} = \gamma_Z G - m_Z Z^2, \quad (136)$$

$$\frac{\partial D}{\partial t} = (1 - \gamma_Z)G + m_P P + m_Z Z^2 - r_D D + w_s \frac{\partial D}{\partial z}, \quad (137)$$

$$\frac{\partial \text{NO}_3}{\partial t} = -\mu_P P + r_D D, \quad (138)$$

where $\mu_P = \mu_{\text{max}} f(T) \min(L_N, L_I)$ with Michaelis-Menten nutrient limitation $L_N = \text{NO}_3/(k_N + \text{NO}_3)$ and Beer-Lambert light limitation with chlorophyll self-shading. Grazing G follows Holling-III. Redfield stoichiometry ($R_{C:N} = 6.625$) couples biology to DIC/ALK.

CMIP Infrastructure

CF/CMOR-Compliant Output

The `CFWriter` class produces CF-1.8 NetCDF4 files with CMIP6 Data Reference Syntax (DRS) filenames: `<var>_<table>_<model>_<expt>_<variant>_<grid>.nc`. Four CMOR tables are provided: `Amon` (33 atmospheric variables including clear-sky radiative fluxes, cloud condensate, and relative humidity), `Lmon` (6 land variables), `Omon` (2 ocean variables: SST and sea-ice fraction), and `Aday` (5 daily atmospheric variables). 3-D fields are interpolated to a standard CMIP6 19-level pressure grid.

AMIP CMIP6 Forcing Deck

The script `scripts/run_amip_cmip6_deck.py` drives the full CMIP6 AMIP protocol with six prescribed-forcing channels: SST/SIC (HadISST/COBE monthly), annual GHG (`greenhouse_historical_pl` schema, year-indexed $\text{CO}_2/\text{CH}_4/\text{N}_2\text{O}/\text{CFC-11}/\text{CFC-12}$; consumed by RRTMG only — gray radiation does not respond), ozone (`vmro3` on (t, ϕ, p) either as monthly climatology or multi-decadal CMIP6 transient via the loader’s non-cyclic dispatch), solar TSI plus 14-band fractions (`swflux_14band_cmip6_*` expanded to 112 RRTMG-SW g -points), aerosol AOD (Kinne zonal monthly), and volcanic AOD (CMIP6 `bc_aeropt_cmip6_volc_*` with 1982/1991 eruptions sampled at correct calendar months). The configs `GHGConfig`, `OzoneConfig`, `SolarConfig`, `AerosolConfig` (`src/legoesm/forcing/external.py`) all carry `start_year` so simulation days map onto file absolute time axes via `_simday_to_file_day`.

The deck physics defaults are RRTMG + Sundqvist cloud fraction + Sundqvist large-scale condensation + SBM + Louis (Kessler microphysics is currently disabled in this path because it produces NaN winds at day ~ 2). When forcing files are absent the user must pass `--auto-generate` to invoke `scripts/generate_amip_forcing.py` for synthetic CMIP6-shape replacements. `scripts/smoke_test` exercises the dispatch path on cubed-sphere (centred / FV3 / C-D), lat-lon (centred / FV3 / C-grid), Gaussian spectral, and Voronoi/MPAS using gray radiation with aerosol/volcanic disabled; it does not validate the full forcing stack on every grid. Voronoi/MPAS additionally requires `--turbulence none` (TKE/MPAS edge-wind mismatch) and `--dt 60` (hidden hydrostatic-dycore CFL). Spectral and MPAS paths bypass `DiagnosticCollector`.

Experiment Templates

Six CMIP6-style experiments with built-in GHG time series:

Template	Period	Forcing	CO ₂ range
<code>piControl</code>	1850– ∞	Fixed 1850	284.3 ppmv
<code>historical</code>	1850–2014	Transient	284→397 ppmv
<code>ssp245</code>	2015–2100	SSP2-4.5	397→600 ppmv
<code>ssp585</code>	2015–2100	SSP5-8.5	397→1135 ppmv
<code>amip</code>	1979–2014	Prescribed SST	Transient
<code>1pctCO2</code>	1–150	1%/yr increase	284→1277 ppmv

Restart and Reproducibility

Each restart file is accompanied by a `.meta.json` sidecar containing: SHA-256 state digest (sorted, concatenated array bytes), SHA-256 config hash, platform identifier, JAX version, x64 mode, and git hash. The function `verify_reproducibility()` compares two restart files and reports any discrepancies.

Tuning Guide

A registry of 16 tunable parameters across five categories (dynamics, radiation, convection, diffusion, surface) with valid ranges, default values, sensitivity ratings, and physical notes. `validate_tuning(config)` checks CFL consistency, range violations, and conflicting options. `recommended_params(resolution, nlev)` provides resolution-appropriate defaults.

OMIP Simulation Pipeline

The `run_omip.py` script provides a complete Ocean Model Intercomparison Project (OMIP) simulation pipeline supporting cubed-sphere, lat-lon, MPAS, and spectral grids. Features include WOA18-based initialisation with fallback, Haney-style SST/SSS restoring toward climatology, configurable physics presets (“none”, “minimal”, “full” with KPP, GM/Redi, bottom drag, shortwave penetration), per-grid resolution and timestep defaults, and diagnostic extraction with wall-time tracking.

4D-Var Data Assimilation (v3.9)

The model includes a fully differentiable 4D-Var data assimilation system (Courtier et al., 1994; Rabier et al., 2000) that exploits JAX’s automatic differentiation to compute exact gradients of the cost function without hand-coded adjoints, replacing the conventional separately-maintained tangent-linear and adjoint models.

Cost Function

The 4D-Var cost function minimised over the assimilation window $[t_0, t_N]$ is

$$J(\mathbf{x}_0) = \frac{1}{2}(\mathbf{x}_0 - \mathbf{x}_b)^T \mathbf{B}^{-1}(\mathbf{x}_0 - \mathbf{x}_b) + \frac{1}{2} \sum_{i=1}^N (\mathcal{H}_i[\mathcal{M}_{0 \rightarrow i}(\mathbf{x}_0)] - \mathbf{y}_i)^T \mathbf{R}_i^{-1}(\mathcal{H}_i[\mathcal{M}_{0 \rightarrow i}(\mathbf{x}_0)] - \mathbf{y}_i), \quad (139)$$

where $\mathbf{x}_b$ is the background state, $\mathbf{B}$ the background error covariance, $\mathcal{M}_{0 \rightarrow i}$ the nonlinear model propagator, $\mathcal{H}_i$ the observation operator at time t_i , $\mathbf{y}_i$ the observations, and $\mathbf{R}_i$ the observation error covariance. Gradients $\nabla_{\mathbf{x}_0} J$ are computed via `jax.grad` through the full forward model trajectory.

Control Vector

The control vector comprises perturbations to:

- Horizontal wind components (u, v),
- Temperature (T),
- Specific humidity (q_v),
- Surface pressure (p_s).

Increments are applied in model space and can optionally be preconditioned by the square root of $\mathbf{B}$ (i.e. $\mathbf{x}_0 = \mathbf{x}_b + \mathbf{B}^{1/2} \boldsymbol{\chi}$ with minimisation in $\boldsymbol{\chi}$ -space).

Background Error Covariance

Four background error covariance formulations are implemented:

1. **Diagonal:** $\mathbf{B} = \text{diag}(\sigma_b^2)$, the simplest variance-only model.
2. **Diffusion:** $\mathbf{B} = \boldsymbol{\Sigma} \mathbf{C} \boldsymbol{\Sigma}$ where $\mathbf{C}$ is a correlation matrix defined implicitly via iterated Laplacian diffusion on the model grid.
3. **Spectral:** $\mathbf{B}$ is diagonal in spectral space, with variance specified per total wavenumber; correlations arise naturally from the spectral truncation.

4. **Hybrid:** $\mathbf{B} = \beta_1 \mathbf{B}_{\text{static}} + \beta_2 \mathbf{B}_{\text{ensemble}}$, blending a static covariance with flow-dependent ensemble covariance, the latter spatially localised by a compactly supported Gaspari and Cohn (1999) correlation function to suppress spurious long-range sampling error.

Observation Operators

Three observation operator types are provided:

- **Direct:** identity mapping from model grid points to observed quantities.
- **Interpolating:** bilinear (horizontal) and linear (vertical) interpolation from the model grid to observation locations.
- **Composite:** chains multiple operators (e.g. interpolation followed by a forward radiative transfer calculation).

Minimiser

The cost function is minimised using preconditioned gradient descent. The preconditioner is the square root of $\mathbf{B}$, transforming the problem to χ -space where the background term becomes isotropic. The minimiser supports configurable iteration counts and convergence tolerances, and all operations are JIT-compiled.

DA Cycling and Incremental Analysis

For operational-style cycling, the analysis from each window serves as the background for the next. An incremental analysis update (IAU) option distributes the analysis increment gradually over the subsequent forecast window rather than inserting it as an instantaneous shock, reducing initialisation imbalance and gravity-wave noise.

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